FLB by Learnability Dimensions — Polymarket — v6

v6 rebuilds every result on the deduplicated clean trades dataset and reports dollar-weighted alongside count-weighted calibration throughout. Versus v5: (1) trades source is the clean parquet (1.377B rows; 58.2M full-row ingestion-replay duplicates removed — see *data_exploration.md*); (2) the bot *wallet_flags* were recomputed on clean data; (3) every spread is reported both count-weighted (each trade = 1 observation) and dollar-weighted (each trade weighted by *usdcSize*); (4) the favorite-longshot slope (D10−D1) is distinguished from a level offset (see "Offset vs slope" below); (5) the count-of-contracts dimensions (family size, outcomes-per-event, group/slug sizes, recurrence, prior-settlements) now count distinct markets (*condition_id*), not outcome tokens — a binary market's YES/NO are separate token_ids that previously double-counted, emptying the "Binary 1"/"Singleton 1" buckets (binaries misfiled into "Few 2-5") and suppressing the "One-off" recurrence class entirely. Counting markets restores those buckets (Binary 1 and One-off are now the strongest-FLB endpoints) and re-cuts the family-size/prior-settlement boundaries onto the correct unit. v5's audit-driven structure is retained: up/down excluded from the primary view (reported separately as *dim_market_type*), fixed-threshold *dim_text_novelty*, three lifecycle windows, Bonferroni-corrected significance, and Appendices A–E.

Setup

Dataset. 1.117M Polymarket contracts (`stage2_per_contract_augmented.parquet`) joined to the clean trade-level parquet (`/mnt/data/pipeline_output/trades_clean.parquet`, 1,377,065,934 rows). The clean set removes 58.2M (4.06%) byte-for-byte full-row duplicates (ingestion-pipeline replays); this changes calibration negligibly (replays are random w.r.t. outcome) but tightens SEs honestly and removes ~2% of penny-trade volume. Dimension assignments (volume tiers, horizons, prior-settlement counts) are reused from the cached contract-dimensions parquet; the trade-derived ones move <2% on clean (volume tiers do not cross boundaries; horizons and prior-settlement counts are timestamp/resolution-based and unchanged).

Trades scope. `side = 'BUY'` only (return-on-stake of the taker). The view excludes (1) wallets flagged non-human by the behavioral bot filter, and (2) up/down markets (`eventSlug NOT LIKE '%updown%' AND NOT LIKE '%up-or-down%'`) — sub-hour noise-trading regimes that contaminated several v3 findings; reported separately as `dim_market_type`.

Weighting (new in v6). Every decile calibration error and every D10−D1 spread is computed two ways: - Count-weighted — `mean(won - price)` over trades; each trade is one observation. The per-decision view (standard FLB convention). - Dollar-weighted — `sum(usdcSize·(won - price)) / sum(usdcSize)`; each trade weighted by stake. The per-dollar (economic-magnitude) view. Tables report both as `spread (t)` pairs. Where they agree, the bias is not a small-trade artifact; where they diverge, trade size carries information (flagged in the headline section). Note: dollar-weighting concentrates the effective sample on fewer large trades, so dollar t-stats are typically a touch lower than count t-stats at equal spread.

Offset vs slope. FLB is a slope: longshots (low deciles) overpriced, favorites (high deciles) underpriced, so D10−D1 > 0. That is distinct from a uniform level offset (buyers beat/lose to the price at every decile), which is a calibration shift, not FLB. v6 reports the D10−D1 spread as the FLB measure; where a slice instead shows a flat nonzero level (notably Sports/Esports, where mid-life buyers win and closing buyers lose), that is documented as an offset in `data_exploration.md` (the Sports/Esports two-regime case study with liquidity-floor diagnostics), not folded into the FLB spread here.

Bot filter (`analysis/learnability/bot_filter.py`). `is_nonhuman` = OR of: A inter-trade-interval median <1s (definite) / 1–10s (likely); B trades-per-active-day >500 (definite) / >200 (likely); C hour-of-day HHI <0.06 AND n>500; E size CV(usdcSize) <0.05 AND n>50. Composite: `A_def OR (A_lik AND any{B,C,E}) OR (B_def AND C) OR (>=2 of {B_lik,C,E})`. Recomputed on clean data: 255,261 wallets flagged (21.20%), 1.11B trades (80.6%) — 458 fewer wallets than the raw-based flags (replays marginally over-flagged). ±25% threshold sensitivity swings headline spreads <=0.005 (audit §11).

Lifecycle windows (reported side-by-side): 25-80% mature (primary), 80-100% closing (secondary), 0-100% full (baseline). Position = `(t.timestamp - mkt_start)/mkt_duration` from min/max trade timestamps. Mechanically incomparable across durations (a <1h crypto's 25-80% is minutes; a >1mo election's is weeks) — see `dim_contract_horizon`.

Inference. 3-way clustered SE on (`trade_day`, `proxyWallet`, `market_id`) via Cameron-Gelbach-Miller (2011) inclusion-exclusion; the dollar-weighted SE uses the weighted-residual analogue (score `e_i = w_i(r_i - theta_w)`, normalizer `(Σw)²`). `market_id` is the per-market `condition_id` so YES/NO cluster together. Unstable on small slices (see Caveats).

Significance. Family ≈ 3 windows × ~22 dims × ~3–5 kept slices × 2 weightings. Bonferroni-equivalent |t|: ∗ > 2.5, ∗∗ > 3.0, ∗∗∗ > 3.5 — not the naive 1.96.

Deciles. Equal-width: D1 = (0.01, 0.10) … D10 = (0.90, 0.99). Minimum slice size 5,000 trades; smaller slices dropped silently.

Reproducibility. `analysis/learnability/{dimensions*.py, flb_per_slice_v3.py, run_phase1_v5.py, bot_filter.py}` + the contract-dimensions parquet + the clean trades parquet on EC2 (`config.py:TRADES_PARQUET_GLOB`).

Headline findings

Strongest favorite-longshot slices (mature 25-80% window, |t|>3.5 count-weighted, `dim_market_type` excluded as a sensitivity slice; sorted by |t|; dollar-weighted alongside):

Dim : slice	N	count spread (t)	dollar spread (t)
`dim_outcomes_per_event` : Few 2-5	4.1M	+0.0474 (+14.9***)	+0.0636 (+13.0***)
`dim_primary_category` : Finance	1.1M	+0.0511 (+12.1***)	+0.0623 (+10.3***)
`dim_text_neighbors_strict` : 0 strict neighbors	2.3M	+0.0556 (+11.9***)	+0.0684 (+6.8***)
`dim_text_novelty` : <0.50 genuinely isolated	2.2M	+0.0556 (+11.8***)	+0.0684 (+6.7***)
`dim_prior_settlements_bin__event_slug` : 6-50	246K	+0.0628 (+11.7***)	+0.0837 (+19.2***)
`dim_family_size_x_vol` : Small 2-20 × High vol	9.0M	+0.0451 (+11.2***)	+0.0583 (+8.1***)
`dim_event_family_size` : Small 2-20	9.2M	+0.0441 (+11.1***)	+0.0580 (+8.0***)
`dim_info_type_supergroup` : other	5.7M	+0.0443 (+10.4***)	+0.0579 (+10.3***)
`dim_prior_settlements_bin__dim_group_strict` : 6-50	228K	+0.0628 (+10.2***)	+0.0802 (+15.7***)
`dim_event_family_size` : Singleton 1	2.6M	+0.0476 (+9.9***)	+0.0567 (+5.9***)
`dim_recurrence_class` : One-off	2.6M	+0.0476 (+9.9***)	+0.0567 (+5.9***)
`dim_family_size_x_vol` : Singleton 1 × High vol	2.4M	+0.0485 (+9.4***)	+0.0572 (+5.8***)
`dim_primary_category` : Geopolitics	1.0M	+0.0510 (+8.8***)	+0.0712 (+13.1***)
`dim_primary_category` : Politics	6.8M	+0.0438 (+8.7***)	+0.0599 (+7.1***)

Dollar vs count — the systematic pattern. Across the 32 mature-window slices significant at |t|>3.0 (count-weighted, non-degenerate), the dollar-weighted spread exceeds the count-weighted spread in 94% of them (median ratio 1.30×). Large-stake trades exhibit stronger favorite-longshot bias than the typical trade — the bias is concentrated in big bets, not diluted by them. This holds across categories (Finance, Politics, Geopolitics), family-size, novelty, and prior-settlement dimensions alike, so the headline FLB findings are economic, not penny-trade artifacts.

No genuine sign flips: wherever both weightings are significant they agree on direction.

Degenerate dollar-weighted slices (|dollar spread| > 0.15) — thin slices where a single very large trade dominates the dollar mean; the dollar SE is not asymptotically valid here, so these are reported for completeness only and excluded from the pattern above:

Window	Dim : slice	N	dollar spread (t)
80-100%	`dim_contract_horizon` : <1h	87K	-0.3406 (-3.7***)
full	`dim_contract_horizon` : <1h	284K	-0.3270 (-5.0***)
25-80%	`dim_dollar_volume_tier` : Q1 (<=$32)	81K	-0.2729 (-1.5)
full	`dim_vol_per_contract_tier` : VPC Q2	540K	-0.1803 (-4.8***)

Window	Dim : slice	N	dollar spread (t)
25-80%	`dim_contract_horizon` : <1h	102K	-0.1736 (-2.6*)
80-100%	`dim_text_neighbors_strict` : 2-5 strict neighbors	16K	-0.1678 (-0.8)
80-100%	`dim_prior_settlements_bin__dim_group_strict` : 1-5	238K	-0.1611 (-1.3)

SE-degenerate slices — the 3-way clustered SE collapses toward zero when a slice has essentially one effective day/wallet/market cluster (e.g. a single recurring event family), producing absurd t-stats. These are excluded from the tables above and reported only as a flag:

Window	Dim : slice	N	spread (t, invalid)
full	`dim_prior_settlements_bin__dim_group_strict` : 50+	71K	+0.0620 (t=+1504)
full	`dim_market_type` : updown	150K	+0.0545 (t=+922)
80-100%	`dim_market_type` : updown	72K	+0.0548 (t=+350)
25-80%	`dim_prior_settlements_bin__dim_group_strict` : 50+	31K	+0.1080 (t=+292)
25-80%	`dim_market_type` : updown	69K	+0.0519 (t=+56)
80-100%	`dim_prior_settlements_bin__dim_group_strict` : 50+	37K	+0.0566 (t=+0)

Calibration grids

Each panel is one dimension; each line is a slice's calibration curve (empirical win rate vs implied probability, by decile). On the gray diagonal = perfectly calibrated; bowed below the diagonal at low prices and above at high prices = favorite-longshot bias (longshots overpriced, favorites underpriced).

Count-weighted (each trade equal):

Dollar-weighted (each trade weighted by `usdcSize`):

Dimension ranking — by mature-window spread variance (count-weighted)

Dim	Spread std	N slices
`dim_dollar_volume_tier`	0.1035	4
`dim_vol_per_contract_tier`	0.0677	5
`dim_vol_per_contract_residualized`	0.0384	3
`dim_prior_settlements_bin__dim_group_strict`	0.0371	4
`dim_event_family_size`	0.0335	4
`dim_family_size_x_vol`	0.0317	9
`dim_prior_settlements_bin__event_template`	0.0307	4
`dim_contract_horizon`	0.0292	5
`dim_recurrence_class`	0.0287	4
`dim_info_type_supergroup`	0.0257	7
`dim_text_novelty`	0.0242	5
`dim_primary_category`	0.0236	12
`dim_market_type`	0.0235	2
`dim_prior_settlements_bin__event_slug`	0.0229	3
`dim_outcomes_per_event`	0.0223	3
`dim_text_neighbors_strict`	0.0217	3
`dim_family_vol_tier`	0.0194	3

Dim	Spread std	N slices
`dim_subject_specificity`	0.0179	3
`dim_group_strict_size`	0.0137	3
`dim_resolution_type`	0.0136	2
`dim_market_specificity`	0.0112	3
`dim_event_slug_size`	0.0102	3

Per-dimension deep dive

1. `dim_resolution_type` — Does the outcome have an objective data feed?

Hypothesis: data-driven outcomes (crypto prices, weather, sports scores) have public reference series -> calibrated; event-observable outcomes (elections, resignations, rulings) -> classic FLB.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
event_observable	16.8M	+0.0250 (+2.7*)	+0.0360 (+2.1)	+0.0306 (+6.6***)	+0.0322 (+3.3**)	+0.0254 (+3.9***)	+0.0292 (+2.6*)
data_driven_numeric	7.3M	+0.0058 (+0.6)	+0.0213 (+2.0)	+0.0158 (+2.2)	+0.0411 (+5.7***)	+0.0060 (+0.9)	+0.0204 (+3.1**)
unknown	—	—	—	—	—	+0.0947 (+16.9***)	+0.1057 (+19.9***)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

event_observable

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.037	0.032	-0.0049	-0.6	-0.0115	-0.7
D2	0.147	0.135	-0.0114	-1.1	-0.0177	-1.0
D3	0.248	0.270	+0.0218	+1.2	-0.0044	-0.2
D4	0.352	0.272	-0.0795	-1.2	-0.1262	-1.7
D5	0.448	0.408	-0.0402	-0.8	-0.0947	-1.0
D6	0.545	0.634	+0.0892	+1.4	+0.1341	+1.7
D7	0.642	0.782	+0.1396	+1.9	+0.1484	+1.9
D8	0.745	0.762	+0.0169	+1.1	+0.0068	+0.3
D9	0.848	0.879	+0.0312	+2.8*	+0.0131	+0.6
D10	0.961	0.981	+0.0201	+6.1***	+0.0245	+8.5***
D10-D1			+0.0250	+2.7*	+0.0360	+2.1

data_driven_numeric

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.042	0.038	-0.0039	-1.2	-0.0131	-1.7
D2	0.143	0.135	-0.0077	-0.9	-0.0224	-1.4
D3	0.246	0.295	+0.0488	+1.1	+0.1237	+1.4
D4	0.346	0.398	+0.0525	+1.2	+0.1704	+1.6
D5	0.452	0.480	+0.0276	+2.2	+0.0454	+1.8
D6	0.536	0.542	+0.0055	+0.3	-0.0441	-1.0
D7	0.646	0.603	-0.0433	-0.6	-0.1802	-1.4
D8	0.743	0.640	-0.1030	-1.0	-0.1532	-1.2
D9	0.849	0.872	+0.0224	+2.6*	+0.0363	+2.6*
D10	0.961	0.963	+0.0019	+0.2	+0.0083	+1.1
D10-D1			+0.0058	+0.6	+0.0213	+2.0

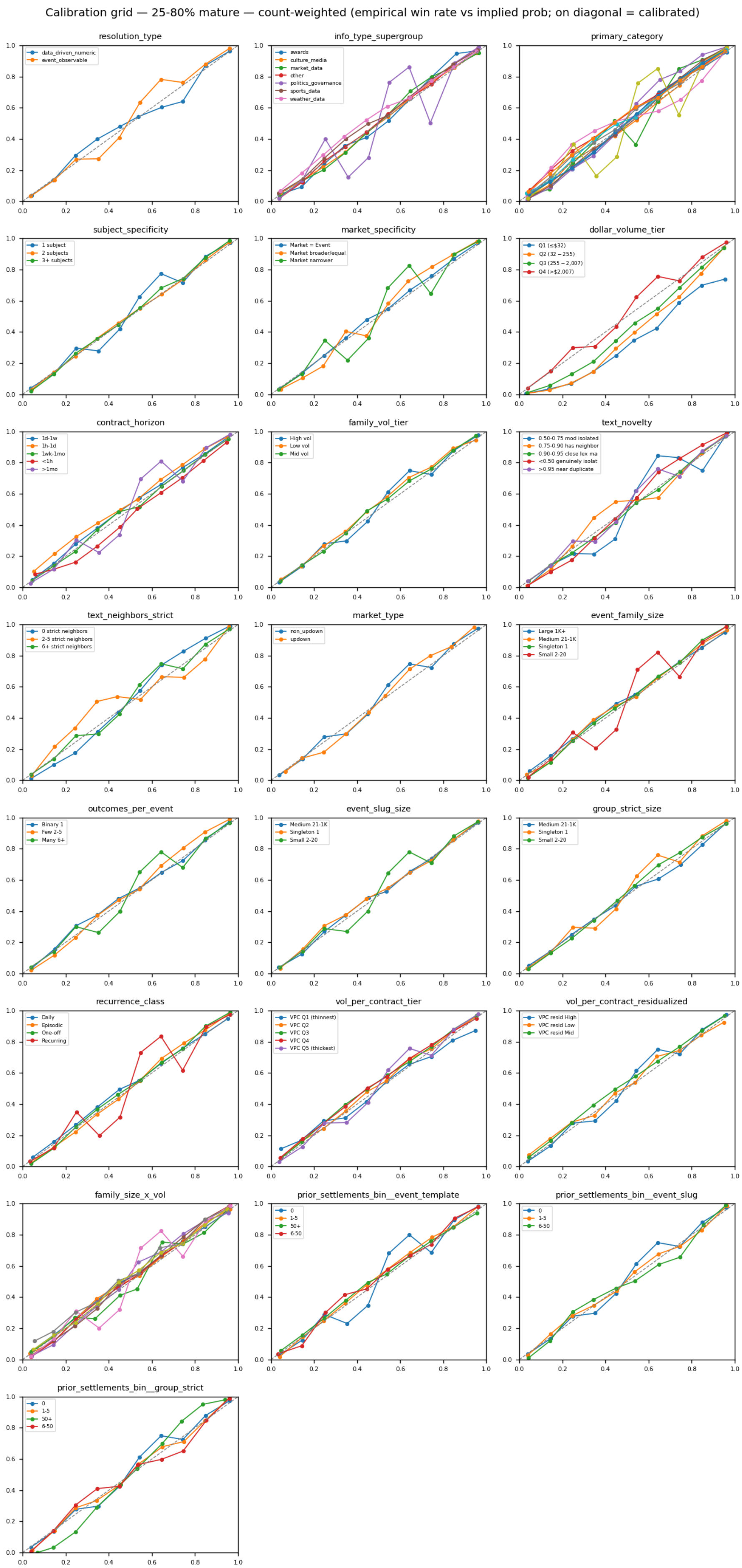

Figure 1: 25-80% mature — count

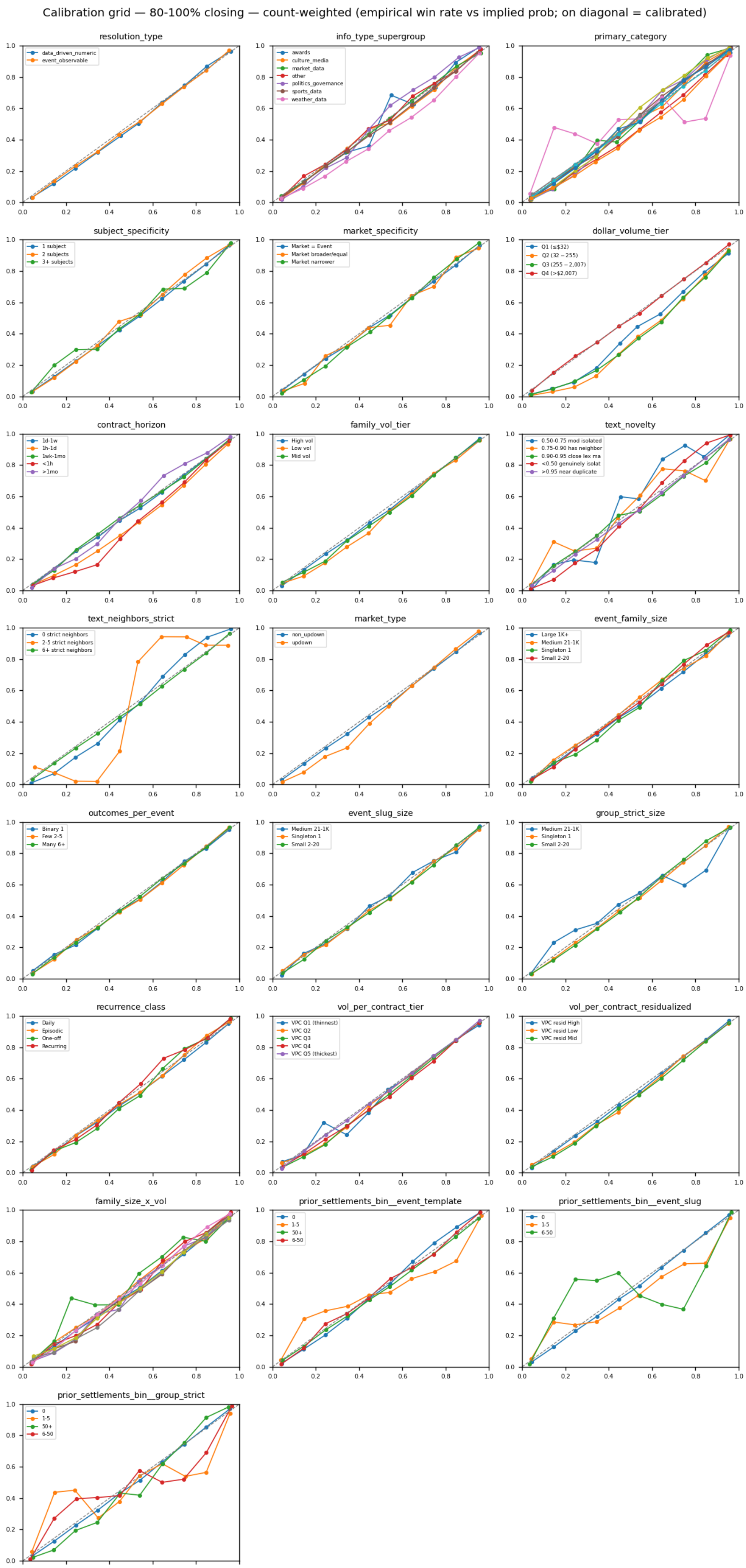

Figure 2: 80-100% closing — count

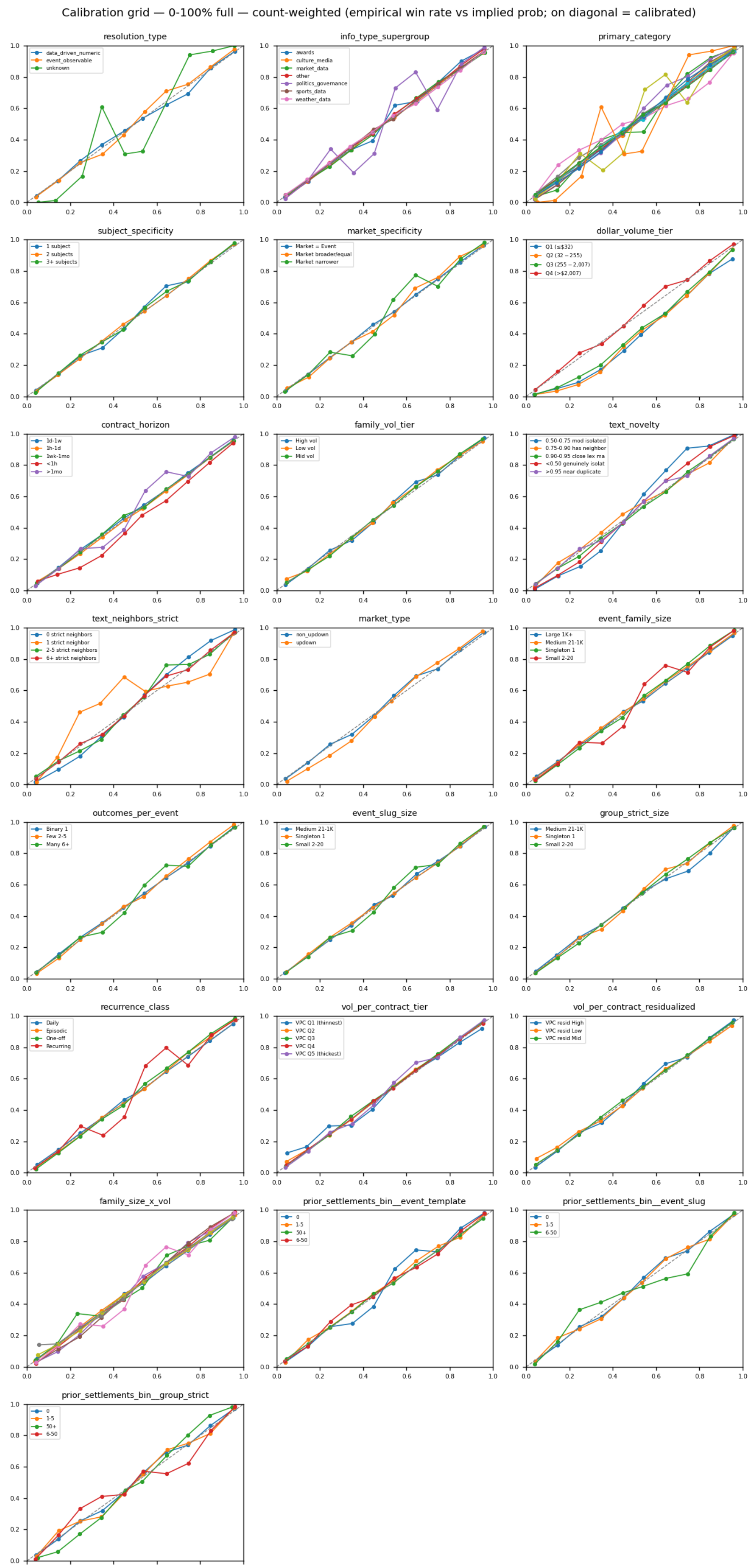

Figure 3: 0-100% full — count

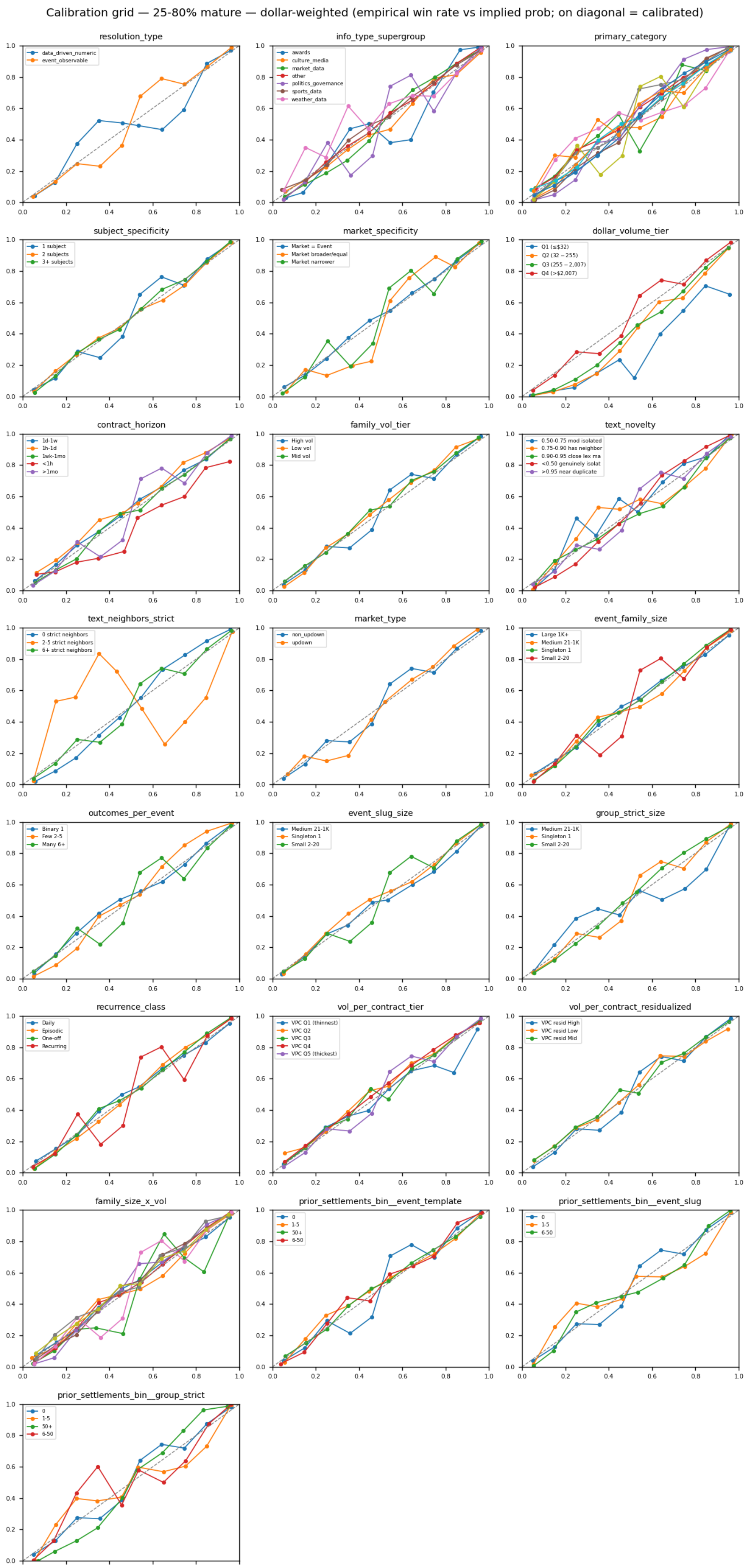

Figure 4: 25-80% mature — dollar

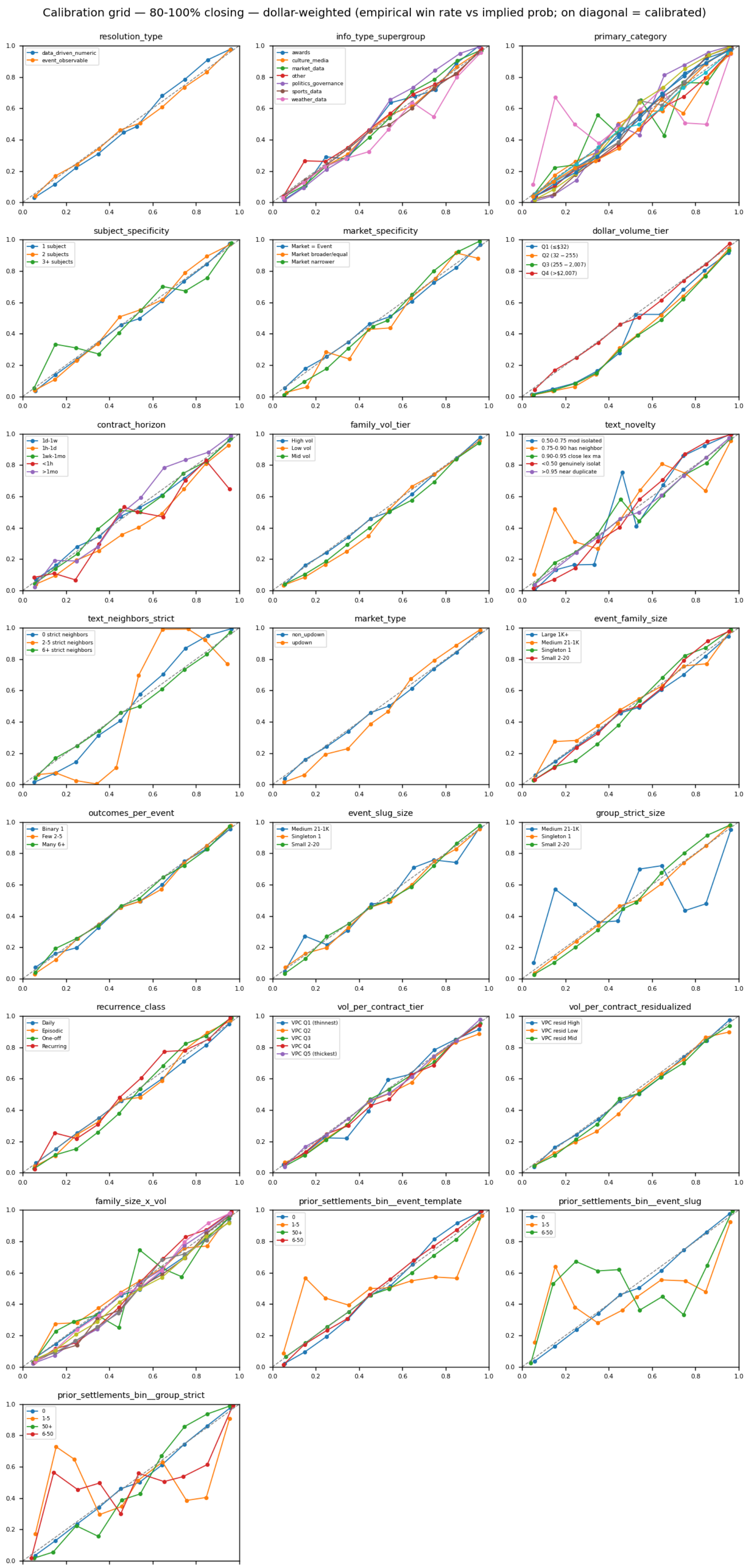

Figure 5: 80-100% closing — dollar

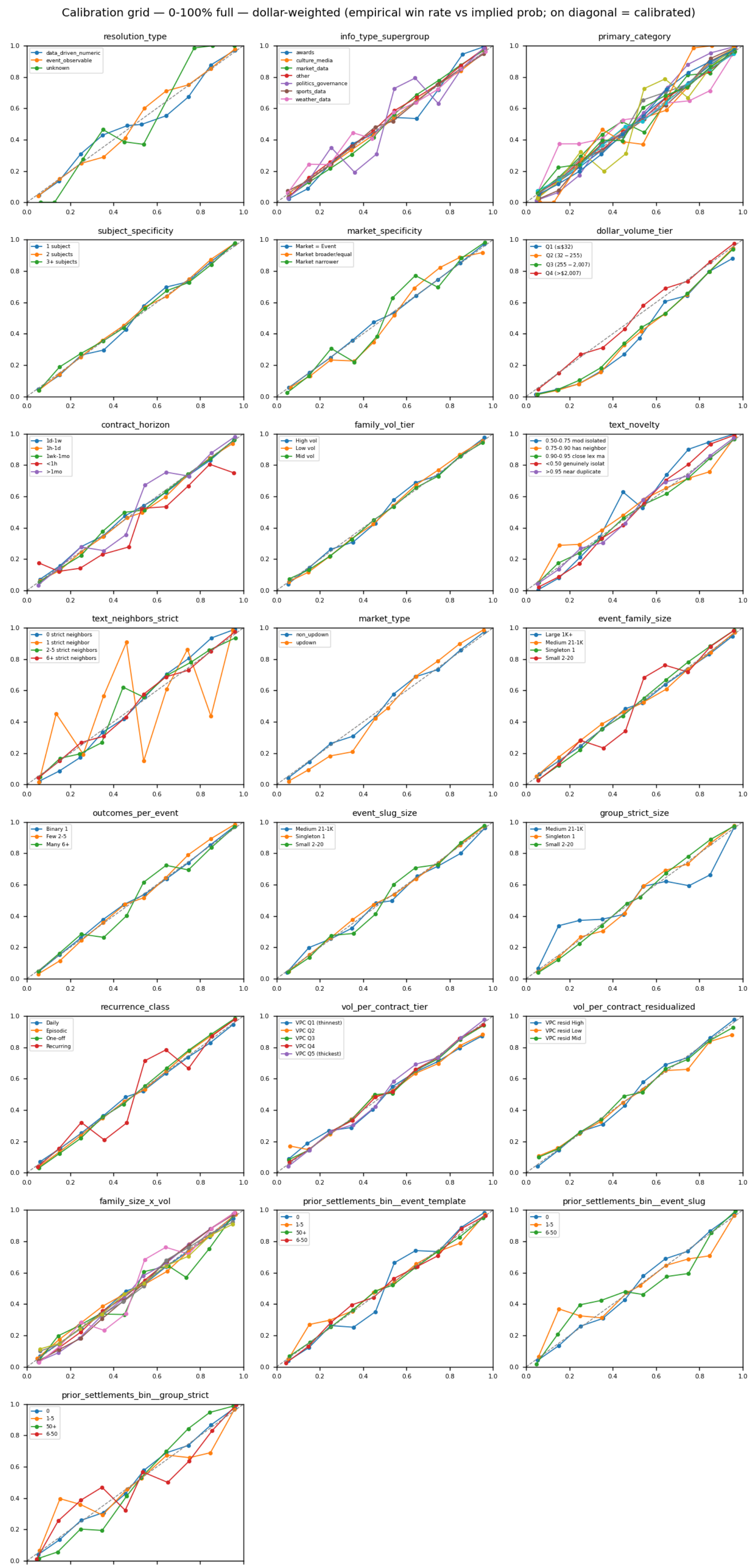

Figure 6: 0-100% full — dollar

2. `dim_info_type_supergroup` — Finer-grained classification of the data source

Hypothesis: market_data / weather_data -> calibrated; politics_governance / awards -> strong FLB.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
sports_data	7.9M	-0.0063	-0.0331	+0.0203	+0.0108	-0.0009	-0.0295
		(-0.3)	(-0.6)	(+6.3***)	(+1.6)	(-0.1)	(-1.0)
politics_governance	5.7M	+0.0456	+0.0581	+0.0550	+0.0817	+0.0476	+0.0544
		(+8.1***)	(+5.6***)	(+5.7***)	(+7.8***)	(+7.4***)	(+5.6***)
other	5.7M	+0.0443	+0.0579	+0.0317	+0.0288	+0.0349	+0.0462
		(+10.4***)	(+10.3***)	(+3.8***)	(+1.6)	(+7.7***)	(+6.0***)
market_data	2.4M	+0.0000	+0.0324	-0.0084	+0.0251	+0.0026	+0.0292
		(+0.0)	(+3.3**)	(-0.5)	(+2.5)	(+0.2)	(+4.1***)
culture_media	1.5M	-0.0132	-0.0164	+0.0300	+0.0227	-0.0027	-0.0053
		(-1.3)	(-1.3)	(+3.4**)	(+1.6)	(-0.4)	(-0.6)
weather_data	673K	-0.0140	-0.0191	+0.0120	+0.0102	-0.0017	-0.0096
		(-2.4)	(-1.1)	(+2.4)	(+1.5)	(-0.4)	(-0.9)
awards	236K	+0.0041	+0.0743	+0.0537	+0.0814	+0.0275	+0.0703
		(+0.1)	(+3.1**)	(+4.8***)	(+9.9***)	(+1.4)	(+5.6***)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

sports_data

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.036	0.051	+0.0157	+0.8	+0.0402	+0.8
D2	0.148	0.145	-0.0035	-0.3	-0.0132	-0.8
D3	0.248	0.277	+0.0288	+1.2	-0.0129	-0.8
D4	0.348	0.400	+0.0519	+3.8***	+0.0453	+1.9
D5	0.450	0.497	+0.0463	+5.2***	+0.0407	+3.1**
D6	0.539	0.548	+0.0091	+1.1	+0.0078	+0.6
D7	0.642	0.668	+0.0261	+2.2	+0.0303	+1.5
D8	0.744	0.779	+0.0348	+2.6*	+0.0220	+0.9
D9	0.846	0.878	+0.0324	+3.7***	+0.0268	+1.5
D10	0.960	0.970	+0.0094	+1.2	+0.0071	+0.8
D10-D1			-0.0063	-0.3	-0.0331	-0.6

politics_governance

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.019	-0.0199	-4.6***	-0.0327	-4.0***
D2	0.144	0.134	-0.0099	-0.4	-0.0197	-0.5
D3	0.252	0.399	+0.1471	+1.7	+0.1265	+1.1
D4	0.358	0.154	-0.2039	-2.3	-0.1886	-1.8
D5	0.452	0.280	-0.1719	-1.8	-0.1649	-1.2
D6	0.549	0.763	+0.2136	+2.1	+0.1972	+2.0
D7	0.641	0.861	+0.2198	+2.1	+0.1720	+1.3
D8	0.740	0.504	-0.2362	-1.7	-0.1644	-1.3
D9	0.852	0.887	+0.0342	+1.0	-0.0207	-0.4
D10	0.963	0.989	+0.0256	+7.2***	+0.0254	+4.0***
D10-D1			+0.0456	+8.1***	+0.0581	+5.6***

other

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.019	-0.0200	-6.9***	-0.0311	-7.2***
D2	0.145	0.123	-0.0223	-1.9	-0.0219	-0.9
D3	0.246	0.256	+0.0100	+0.4	+0.0102	+0.3
D4	0.345	0.349	+0.0040	+0.2	+0.0118	+0.4
D5	0.444	0.443	-0.0005	-0.0	-0.0029	-0.1
D6	0.544	0.559	+0.0155	+1.3	+0.0266	+1.1
D7	0.647	0.666	+0.0193	+0.8	+0.0056	+0.1
D8	0.746	0.750	+0.0044	+0.2	+0.0099	+0.2
D9	0.848	0.879	+0.0306	+2.0	+0.0357	+1.2
D10	0.961	0.985	+0.0243	+7.8***	+0.0268	+7.5***
D10-D1			+0.0443	+10.4***	+0.0579	+10.3***

3. `dim_primary_category` — Polymarket's 13-category taxonomy

Hypothesis: high-information categories calibrated; thin/one-off categories (Politics, Geopolitics, Finance) strong FLB.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
Sports	6.9M	-0.0064 (-0.3)	-0.0343 (-0.6)	+0.0217 (+6.4***)	+0.0101 (+1.3)	-0.0002 (-0.0)	-0.0323 (-1.0)
Politics	6.8M	+0.0438 (+8.7***)	+0.0599 (+7.1***)	+0.0518 (+6.5***)	+0.0748 (+7.5***)	+0.0454 (+8.2***)	+0.0563 (+7.2***)
Crypto	2.7M	+0.0030 (+0.2)	+0.0337 (+3.7***)	-0.0017 (-0.1)	+0.0283 (+2.6*)	+0.0063 (+0.5)	+0.0337 (+5.1***)
Tech	1.7M	+0.0128 (+1.5)	+0.0236 (+2.5*)	+0.0359 (+4.4***)	+0.0373 (+3.1**)	+0.0078 (+1.1)	+0.0184 (+2.4)
Finance	1.1M	+0.0511 (+12.1***)	+0.0623 (+10.3***)	+0.0563 (+9.1***)	+0.0773 (+12.2***)	+0.0483 (+10.8***)	+0.0642 (+12.2***)
Geopolitics	1.0M	+0.0510 (+8.8***)	+0.0712 (+13.1***)	+0.0462 (+5.6***)	+0.0621 (+4.6***)	+0.0443 (+7.6***)	+0.0635 (+9.6***)
Esports	994K	-0.0080 (-0.6)	-0.0142 (-0.6)	+0.0132 (+1.5)	+0.0151 (+1.3)	-0.0068 (-0.5)	-0.0042 (-0.4)
Culture	896K	+0.0443 (+6.5***)	+0.0577 (+7.5***)	+0.0445 (+6.2***)	+0.0511 (+4.9***)	+0.0329 (+6.1***)	+0.0365 (+4.3***)
Iran	851K	+0.0084 (+0.3)	+0.0244 (+0.8)	-0.0441 (-1.2)	-0.0772 (-1.3)	-0.0037 (-0.2)	-0.0159 (-0.5)
Weather	743K	-0.0130 (-2.0)	-0.0106 (-0.7)	+0.0107 (+2.2)	+0.0013 (+0.1)	-0.0014 (-0.3)	-0.0095 (-0.8)
Mentions	311K	+0.0219 (+2.4)	+0.0573 (+6.1***)	+0.0153 (+1.9)	+0.0249 (+2.4)	+0.0079 (+1.0)	+0.0272 (+3.2**)
Economy	138K	+0.0217 (+1.4)	-0.0015 (-0.0)	+0.0306 (+2.3)	+0.0260 (+1.1)	+0.0239 (+2.1)	+0.0182 (+0.5)
Uncategorized	—	—	—	—	—	+0.0947 (+16.9***)	+0.1057 (+19.9***)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

Sports

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.035	0.050	+0.0155	+0.8	+0.0410	+0.8
D2	0.149	0.136	-0.0122	-0.8	-0.0162	-0.8
D3	0.248	0.268	+0.0198	+0.7	-0.0294	-1.6
D4	0.348	0.399	+0.0506	+3.1**	+0.0459	+1.7
D5	0.451	0.495	+0.0444	+4.4***	+0.0432	+2.9*

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D6	0.538	0.540	+0.0018	+0.2	+0.0026	+0.2
D7	0.642	0.665	+0.0223	+1.6	+0.0257	+1.1
D8	0.744	0.777	+0.0332	+2.1	+0.0160	+0.5
D9	0.847	0.881	+0.0349	+3.6***	+0.0281	+1.4
D10	0.961	0.971	+0.0091	+1.1	+0.0067	+0.7
D10-D1			-0.0064	-0.3	-0.0343	-0.6

Politics

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.019	-0.0194	-5.1***	-0.0328	-4.6***
D2	0.144	0.131	-0.0132	-0.6	-0.0182	-0.6
D3	0.251	0.365	+0.1136	+1.5	+0.1084	+1.1
D4	0.357	0.164	-0.1938	-2.2	-0.1848	-1.8
D5	0.452	0.284	-0.1675	-1.8	-0.1648	-1.2
D6	0.549	0.756	+0.2071	+2.0	+0.1952	+2.0
D7	0.641	0.851	+0.2092	+2.0	+0.1636	+1.3
D8	0.741	0.555	-0.1858	-1.5	-0.1363	-1.3
D9	0.851	0.885	+0.0339	+1.3	-0.0063	-0.1
D10	0.964	0.988	+0.0244	+7.4***	+0.0272	+6.0***
D10-D1			+0.0438	+8.7***	+0.0599	+7.1***

Crypto

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.040	0.029	-0.0112	-2.5	-0.0197	-2.8*
D2	0.145	0.138	-0.0074	-0.6	-0.0153	-0.7
D3	0.245	0.205	-0.0397	-2.8*	-0.0498	-2.0
D4	0.345	0.310	-0.0346	-1.9	-0.0473	-1.6
D5	0.445	0.434	-0.0111	-0.7	-0.0380	-1.7
D6	0.543	0.553	+0.0101	+0.8	+0.0224	+1.2
D7	0.647	0.698	+0.0509	+1.9	+0.0478	+1.6
D8	0.746	0.785	+0.0391	+1.8	+0.0281	+1.2
D9	0.849	0.856	+0.0074	+0.5	+0.0226	+1.2
D10	0.963	0.954	-0.0083	-0.5	+0.0140	+2.5*
D10-D1			+0.0030	+0.2	+0.0337	+3.7***

4. `dim_subject_specificity` — How many entities does the event resolve on?

Hypothesis: more subjects = compound = harder to price = stronger FLB.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
1 subject	15.3M	+0.0074 (+0.6)	+0.0248 (+1.2)	+0.0233 (+4.6***)	+0.0431 (+6.3***)	+0.0134 (+1.6)	+0.0220 (+1.7)
2 subjects	6.0M	+0.0272 (+4.8***)	+0.0399 (+4.9***)	+0.0275 (+3.5***)	+0.0275 (+2.3)	+0.0216 (+4.5***)	+0.0326 (+4.5***)
3+ subjects	2.8M	+0.0431 (+7.8***)	+0.0590 (+8.4***)	+0.0315 (+3.2**)	+0.0134 (+0.4)	+0.0362 (+6.7***)	+0.0367 (+2.8*)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

1 subject

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.037	0.040	+0.0031	+0.3	-0.0050	-0.2
D2	0.146	0.133	-0.0127	-1.3	-0.0347	-1.9
D3	0.248	0.296	+0.0478	+1.6	+0.0339	+0.7
D4	0.352	0.278	-0.0738	-1.1	-0.1102	-1.2
D5	0.451	0.419	-0.0322	-0.6	-0.0788	-0.8
D6	0.542	0.625	+0.0829	+1.3	+0.1101	+1.4
D7	0.642	0.774	+0.1315	+1.6	+0.1220	+1.3
D8	0.743	0.715	-0.0279	-0.5	-0.0354	-0.6
D9	0.848	0.885	+0.0364	+3.2**	+0.0264	+1.1
D10	0.962	0.972	+0.0105	+1.6	+0.0198	+5.4***
D10-D1			+0.0074	+0.6	+0.0248	+1.2

2 subjects

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.042	0.030	-0.0117	-3.5***	-0.0201	-3.4**
D2	0.144	0.140	-0.0033	-0.3	+0.0120	+0.6
D3	0.245	0.245	+0.0000	+0.0	+0.0195	+0.5
D4	0.345	0.359	+0.0133	+0.7	+0.0235	+0.7
D5	0.444	0.457	+0.0132	+0.6	-0.0109	-0.3
D6	0.544	0.552	+0.0081	+0.5	+0.0113	+0.5
D7	0.646	0.642	-0.0038	-0.2	-0.0320	-0.7
D8	0.746	0.738	-0.0081	-0.4	-0.0382	-0.9
D9	0.849	0.862	+0.0131	+1.0	+0.0057	+0.3
D10	0.960	0.976	+0.0155	+3.4**	+0.0198	+3.5***
D10-D1			+0.0272	+4.8***	+0.0399	+4.9***

3+ subjects

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.038	0.020	-0.0180	-3.8***	-0.0296	-4.7***
D2	0.146	0.129	-0.0162	-0.8	-0.0198	-0.5
D3	0.246	0.263	+0.0167	+0.5	+0.0264	+0.4
D4	0.347	0.356	+0.0090	+0.3	+0.0112	+0.3
D5	0.443	0.444	+0.0005	+0.0	-0.0191	-0.6
D6	0.545	0.554	+0.0085	+0.4	+0.0131	+0.5
D7	0.644	0.683	+0.0386	+1.4	+0.0398	+1.0
D8	0.745	0.743	-0.0019	-0.0	-0.0032	-0.0
D9	0.848	0.877	+0.0288	+1.3	+0.0114	+0.2
D10	0.960	0.985	+0.0251	+8.5***	+0.0294	+9.7***
D10-D1			+0.0431	+7.8***	+0.0590	+8.4***

5. `dim_event_family_size` — How many contracts share the event template?

Hypothesis: large families (NBA games, daily Fed contracts) -> repeated-play learning -> calibrated; singletons/small families -> FLB.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
Small 2-20	9.2M	+0.0441 (+11.1***)	+0.0580 (+8.0***)	+0.0304 (+3.6***)	+0.0492 (+3.8***)	+0.0368 (+6.6***)	+0.0507 (+7.3***)
Large 1K+	6.6M	-0.0193 (-2.1)	-0.0152 (-1.6)	+0.0064 (+1.6)	-0.0053 (-0.7)	-0.0081 (-1.7)	-0.0132 (-2.5)

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
Medium	5.7M	-0.0025	-0.0009	+0.0331	+0.0398	+0.0061	+0.0030
21-1K		(-0.1)	(-0.0)	(+3.8***)	(+2.3)	(+0.5)	(+0.1)
Singleton	2.6M	+0.0476	+0.0567	+0.0456	+0.0523	+0.0428	+0.0493
1		(+9.9***)	(+5.9***)	(+8.2***)	(+5.6***)	(+10.8***)	(+6.5***)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

Small 2-20

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.041	0.022	-0.0188	-6.7***	-0.0327	-5.4***
D2	0.146	0.135	-0.0108	-0.7	-0.0179	-0.7
D3	0.247	0.308	+0.0608	+1.3	+0.0615	+0.9
D4	0.355	0.206	-0.1490	-1.8	-0.1715	-1.8
D5	0.450	0.327	-0.1233	-1.6	-0.1511	-1.2
D6	0.548	0.710	+0.1619	+1.7	+0.1843	+2.0
D7	0.642	0.822	+0.1794	+1.8	+0.1646	+1.5
D8	0.744	0.665	-0.0790	-0.9	-0.0731	-0.8
D9	0.849	0.887	+0.0375	+2.3	+0.0237	+0.8
D10	0.962	0.987	+0.0253	+9.0***	+0.0253	+6.4***
D10-D1			+0.0441	+11.1***	+0.0580	+8.0***

Large 1K+

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.047	0.060	+0.0129	+2.6*	+0.0131	+2.0
D2	0.145	0.158	+0.0126	+1.4	+0.0005	+0.0
D3	0.246	0.267	+0.0201	+2.3	-0.0136	-0.8
D4	0.348	0.381	+0.0331	+3.2**	+0.0299	+1.2
D5	0.450	0.493	+0.0433	+5.1***	+0.0398	+3.0*
D6	0.538	0.551	+0.0129	+1.8	+0.0158	+1.2
D7	0.642	0.661	+0.0184	+1.9	+0.0221	+1.2
D8	0.743	0.761	+0.0183	+2.0	+0.0094	+0.3
D9	0.846	0.850	+0.0046	+0.5	-0.0164	-0.9
D10	0.954	0.947	-0.0064	-0.8	-0.0021	-0.3
D10-D1			-0.0193	-2.1	-0.0152	-1.6

Medium 21-1K

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.034	0.039	+0.0047	+0.3	+0.0163	+0.5
D2	0.144	0.114	-0.0301	-2.3	-0.0306	-1.1
D3	0.248	0.261	+0.0138	+0.3	+0.0270	+0.6
D4	0.344	0.390	+0.0459	+1.7	+0.0807	+2.1
D5	0.447	0.478	+0.0311	+1.5	+0.0145	+0.4
D6	0.541	0.536	-0.0056	-0.3	-0.0464	-1.4
D7	0.646	0.661	+0.0149	+0.6	-0.0656	-1.2
D8	0.746	0.761	+0.0143	+0.5	-0.0233	-0.4
D9	0.850	0.879	+0.0288	+1.7	+0.0218	+0.6
D10	0.963	0.965	+0.0022	+0.2	+0.0154	+3.0**
D10-D1			-0.0025	-0.1	-0.0009	-0.0

6. `dim_outcomes_per_event` — Distinct markets (condition_id) per event_slug (binary = 1 market)

Hypothesis: binary events easiest to calibrate; multi-outcome hardest.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
Many 6+	14.8M	+0.0027 (+0.3)	+0.0186 (+0.9)	+0.0240 (+4.1***)	+0.0345 (+2.5*)	+0.0068 (+1.0)	+0.0141 (+1.0)
Few 2-5	4.1M	+0.0474 (+14.9***)	+0.0636 (+13.0***)	+0.0247 (+5.5***)	+0.0429 (+5.2***)	+0.0380 (+5.6***)	+0.0472 (+7.7***)
Binary 1	3.0M	+0.0239 (+2.5)	+0.0386 (+3.6***)	-0.0070 (-0.5)	-0.0146 (-0.9)	+0.0157 (+1.9)	+0.0220 (+2.3)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

Many 6+

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.041	+0.0023	+0.3	-0.0029	-0.1
D2	0.144	0.140	-0.0044	-0.5	-0.0048	-0.2
D3	0.247	0.300	+0.0523	+1.7	+0.0691	+1.2
D4	0.352	0.262	-0.0906	-1.2	-0.1400	-1.5
D5	0.452	0.398	-0.0539	-0.9	-0.1071	-1.0
D6	0.543	0.650	+0.1068	+1.5	+0.1368	+1.6
D7	0.642	0.780	+0.1381	+1.5	+0.1300	+1.1
D8	0.743	0.678	-0.0646	-1.1	-0.1065	-1.6
D9	0.850	0.867	+0.0166	+1.2	-0.0184	-0.6
D10	0.961	0.966	+0.0051	+0.9	+0.0157	+3.4**
D10-D1			+0.0027	+0.3	+0.0186	+0.9

Few 2-5

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.040	0.021	-0.0189	-7.3***	-0.0355	-7.8***
D2	0.148	0.118	-0.0300	-3.4**	-0.0658	-4.6***
D3	0.247	0.231	-0.0159	-1.0	-0.0572	-1.7
D4	0.347	0.372	+0.0247	+1.3	+0.0462	+1.2
D5	0.448	0.470	+0.0226	+1.4	+0.0202	+0.8
D6	0.541	0.541	-0.0006	-0.0	-0.0050	-0.2
D7	0.643	0.691	+0.0484	+3.0*	+0.0725	+2.2
D8	0.746	0.805	+0.0591	+4.1***	+0.1047	+3.4**
D9	0.846	0.907	+0.0608	+7.1***	+0.0899	+8.6***
D10	0.961	0.990	+0.0284	+15.8***	+0.0281	+15.5***
D10-D1			+0.0474	+14.9***	+0.0636	+13.0***

Binary 1

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.043	0.034	-0.0089	-1.4	-0.0183	-2.2
D2	0.149	0.159	+0.0096	+0.6	+0.0038	+0.1
D3	0.247	0.306	+0.0584	+2.1	+0.0419	+1.1
D4	0.347	0.375	+0.0277	+1.3	+0.0648	+2.0
D5	0.444	0.479	+0.0352	+1.9	+0.0562	+2.7*
D6	0.544	0.549	+0.0052	+0.5	+0.0138	+0.6

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D7	0.645	0.648	+0.0031	+0.1	-0.0258	-0.6
D8	0.744	0.724	-0.0199	-0.6	-0.0186	-0.4
D9	0.845	0.855	+0.0093	+0.5	+0.0167	+0.7
D10	0.962	0.977	+0.0150	+2.1	+0.0203	+3.2**
D10-D1			+0.0239	+2.5	+0.0386	+3.6***

7. `dim_market_specificity` — Does the specific market narrow its parent event?

Hypothesis: narrow markets within broad events -> FLB on the longshot legs.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
Market = Event	13.9M	+0.0154 (+1.8)	+0.0115 (+0.4)	+0.0138 (+2.7*)	+0.0119 (+1.2)	+0.0128 (+2.5)	+0.0133 (+1.1)
Market narrower	10.0M	+0.0234 (+2.3)	+0.0528 (+6.9***)	+0.0497 (+10.5***)	+0.0775 (+11.7***)	+0.0284 (+3.5**)	+0.0463 (+4.6***)
Market broader/equal	139K	+0.0376 (+3.2**)	+0.0572 (+3.8***)	+0.0089 (+0.6)	-0.0297 (-0.5)	+0.0032 (+0.2)	-0.0273 (-0.8)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

Market = Event

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.042	0.037	-0.0048	-0.7	+0.0066	+0.3
D2	0.146	0.137	-0.0082	-1.2	-0.0158	-1.2
D3	0.246	0.249	+0.0028	+0.3	-0.0084	-0.5
D4	0.347	0.361	+0.0144	+1.6	+0.0239	+1.4
D5	0.445	0.479	+0.0344	+4.5***	+0.0361	+2.7*
D6	0.543	0.548	+0.0045	+0.6	+0.0047	+0.4
D7	0.644	0.668	+0.0233	+2.3	+0.0155	+0.9
D8	0.745	0.759	+0.0142	+1.1	+0.0018	+0.1
D9	0.847	0.868	+0.0202	+2.4	+0.0152	+0.9
D10	0.960	0.970	+0.0106	+2.0	+0.0181	+4.3***
D10-D1			+0.0154	+1.8	+0.0115	+0.4

Market narrower

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.035	0.031	-0.0042	-0.5	-0.0278	-4.2***
D2	0.144	0.132	-0.0125	-0.8	-0.0250	-0.8
D3	0.250	0.346	+0.0955	+1.8	+0.0981	+1.1
D4	0.355	0.219	-0.1365	-1.5	-0.1688	-1.6
D5	0.455	0.362	-0.0926	-1.3	-0.1256	-1.1
D6	0.542	0.682	+0.1396	+1.6	+0.1528	+1.7
D7	0.642	0.827	+0.1849	+1.8	+0.1633	+1.3
D8	0.742	0.645	-0.0966	-1.0	-0.0900	-0.9
D9	0.851	0.898	+0.0464	+2.6*	+0.0245	+0.6
D10	0.963	0.983	+0.0192	+4.1***	+0.0250	+7.0***
D10-D1			+0.0234	+2.3	+0.0528	+6.9***

Market broader/equal

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.045	0.033	-0.0120	-1.3	-0.0313	-2.5*
D2	0.148	0.106	-0.0418	-1.5	+0.0209	+0.3
D3	0.242	0.181	-0.0612	-1.6	-0.1171	-2.6*
D4	0.347	0.403	+0.0566	+0.9	-0.1720	-1.5
D5	0.443	0.375	-0.0688	-1.3	-0.2324	-2.5
D6	0.545	0.582	+0.0375	+0.5	+0.0683	+0.6
D7	0.637	0.725	+0.0879	+1.5	+0.1247	+1.9
D8	0.746	0.817	+0.0707	+2.0	+0.1367	+3.7***
D9	0.842	0.894	+0.0514	+1.4	-0.0181	-0.2
D10	0.955	0.981	+0.0256	+3.7***	+0.0259	+2.9*
D10-D1			+0.0376	+3.2**	+0.0572	+3.8***

8. `dim_dollar_volume_tier` — Quartiles of per-token dollar volume

Hypothesis: high-volume contracts attract pros + arbitrage -> calibrated; thin contracts -> retail-only -> FLB.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
Q4 (>$2,007)	22.0M	+0.0137 (+1.7)	+0.0316 (+2.3)	+0.0210 (+4.7***)	+0.0327 (+4.0***)	+0.0130 (+2.3)	+0.0252 (+2.9*)
Q3 ($255-$2,007)	1.5M	+0.0162 (+4.5***)	+0.0381 (+13.3***)	+0.0036 (+1.0)	+0.0176 (+4.9***)	+0.0121 (+4.8***)	+0.0241 (+9.2***)
Q2 ($32-$255)	449K	+0.0236 (+4.8***)	+0.0350 (+6.1***)	+0.0159 (+4.4***)	+0.0289 (+7.4***)	+0.0143 (+2.8*)	+0.0291 (+8.2***)
Q1 (<=$32)	81K	-0.1891 (-1.4)	-0.2729 (-1.5)	-0.0155 (-1.6)	-0.0060 (-0.5)	-0.0524 (-3.0**)	-0.0431 (-2.1)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

Q4 (>$2,007)

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.040	+0.0010	+0.1	-0.0105	-0.8
D2	0.146	0.150	+0.0039	+0.5	-0.0162	-1.1
D3	0.248	0.299	+0.0512	+2.4	+0.0331	+0.9
D4	0.351	0.307	-0.0439	-0.7	-0.0831	-1.1
D5	0.449	0.432	-0.0172	-0.4	-0.0708	-0.8
D6	0.543	0.623	+0.0796	+1.5	+0.1007	+1.3
D7	0.643	0.756	+0.1130	+1.6	+0.1006	+1.2
D8	0.744	0.727	-0.0170	-0.5	-0.0309	-0.7
D9	0.849	0.880	+0.0313	+3.6***	+0.0187	+1.0
D10	0.961	0.976	+0.0146	+3.9***	+0.0211	+7.3***
D10-D1			+0.0137	+1.7	+0.0316	+2.3

Q3 ($255-$2,007)

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.037	0.010	-0.0270	-30.8***	-0.0409	-40.7***
D2	0.143	0.058	-0.0853	-31.4***	-0.1049	-43.2***
D3	0.244	0.130	-0.1134	-22.5***	-0.1361	-29.2***
D4	0.344	0.210	-0.1340	-22.0***	-0.1462	-17.0***
D5	0.448	0.343	-0.1047	-15.5***	-0.1087	-12.7***
D6	0.536	0.457	-0.0792	-11.4***	-0.0766	-7.8***

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D7	0.643	0.550	-0.0930	-12.4***	-0.1016	-10.5***
D8	0.743	0.683	-0.0599	-8.4***	-0.0709	-8.6***
D9	0.845	0.813	-0.0322	-5.0***	-0.0245	-3.8***
D10	0.951	0.940	-0.0108	-3.1**	-0.0027	-1.0
D10-D1			+0.0162	+4.5***	+0.0381	+13.3***

Q2 ($32-$255)

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.006	-0.0328	-20.1***	-0.0420	-48.4***
D2	0.140	0.027	-0.1123	-48.5***	-0.1142	-42.9***
D3	0.241	0.072	-0.1688	-43.1***	-0.1670	-34.4***
D4	0.342	0.143	-0.1991	-31.2***	-0.1979	-29.3***
D5	0.448	0.294	-0.1546	-14.7***	-0.1608	-14.2***
D6	0.534	0.396	-0.1382	-9.4***	-0.0951	-2.2
D7	0.637	0.515	-0.1222	-3.5**	-0.0281	-0.3
D8	0.741	0.623	-0.1180	-9.0***	-0.1132	-8.5***
D9	0.843	0.775	-0.0680	-5.8***	-0.0580	-5.1***
D10	0.951	0.942	-0.0091	-2.0	-0.0070	-1.2
D10-D1			+0.0236	+4.8***	+0.0350	+6.1***

9. `dim_contract_horizon` — Duration from first to last trade on the contract

Hypothesis: medium-horizon best calibrated; ultra-short and ultra-long both worse.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
>1mo	13.4M	+0.0272	+0.0385	+0.0456	+0.0588	+0.0294	+0.0381
		(+2.9*)	(+2.3)	(+6.1***)	(+4.8***)	(+4.1***)	(+3.4**)
1wk-1mo	4.8M	-0.0025	+0.0158	+0.0158	+0.0185	+0.0003	+0.0083
		(-0.3)	(+1.8)	(+2.8*)	(+2.0)	(+0.1)	(+1.1)
1d-1w	4.0M	+0.0047	+0.0080	+0.0088	-0.0041	+0.0044	-0.0071
		(+0.8)	(+0.8)	(+2.4)	(-0.5)	(+1.2)	(-1.1)
1h-1d	1.8M	-0.0348	-0.0355	-0.0012	-0.0040	-0.0098	-0.0149
		(-4.8***)	(-2.6*)	(-0.3)	(-0.4)	(-2.2)	(-1.9)
<1h	102K	-0.0437	-0.1736	+0.0151	-0.3406	-0.0225	-0.3270
		(-1.2)	(-2.6*)	(+0.9)	(-3.7***)	(-1.1)	(-5.0***)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

>1mo

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.036	0.027	-0.0089	-1.1	-0.0148	-0.9
D2	0.145	0.118	-0.0270	-2.0	-0.0253	-1.3
D3	0.249	0.304	+0.0544	+1.3	+0.0589	+1.0
D4	0.354	0.223	-0.1309	-1.6	-0.1438	-1.5
D5	0.450	0.338	-0.1120	-1.5	-0.1391	-1.1
D6	0.548	0.694	+0.1467	+1.5	+0.1678	+1.7
D7	0.643	0.810	+0.1671	+1.8	+0.1396	+1.3
D8	0.744	0.681	-0.0626	-0.9	-0.0624	-0.9
D9	0.851	0.895	+0.0442	+3.0**	+0.0271	+1.1
D10	0.964	0.982	+0.0183	+3.9***	+0.0237	+6.9***

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D10-D1			+0.0272	+2.9*	+0.0385	+2.3

1wk-1mo

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.044	0.044	-0.0000	-0.0	-0.0066	-1.0
D2	0.145	0.138	-0.0071	-0.7	-0.0216	-1.8
D3	0.244	0.230	-0.0148	-1.2	-0.0465	-2.8*
D4	0.346	0.368	+0.0228	+1.4	+0.0251	+0.8
D5	0.445	0.484	+0.0391	+2.6*	+0.0391	+1.3
D6	0.543	0.518	-0.0249	-1.5	-0.0288	-1.1
D7	0.644	0.648	+0.0039	+0.3	+0.0087	+0.4
D8	0.746	0.749	+0.0024	+0.2	-0.0067	-0.2
D9	0.847	0.853	+0.0057	+0.5	-0.0022	-0.1
D10	0.956	0.953	-0.0025	-0.3	+0.0092	+1.7
D10-D1			-0.0025	-0.3	+0.0158	+1.8

1d-1w

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.045	0.049	+0.0049	+1.4	+0.0048	+0.6
D2	0.146	0.155	+0.0090	+1.2	+0.0129	+0.6
D3	0.246	0.276	+0.0306	+4.2***	+0.0373	+2.2
D4	0.347	0.380	+0.0332	+3.6***	+0.0250	+0.9
D5	0.447	0.484	+0.0368	+3.0*	+0.0239	+1.2
D6	0.541	0.574	+0.0328	+3.6***	+0.0421	+2.1
D7	0.643	0.660	+0.0170	+1.6	+0.0185	+0.8
D8	0.743	0.768	+0.0252	+3.1**	+0.0233	+1.2
D9	0.846	0.856	+0.0102	+1.1	-0.0090	-0.4
D10	0.953	0.963	+0.0095	+2.0	+0.0128	+3.0**
D10-D1			+0.0047	+0.8	+0.0080	+0.8

10. `dim_recurrence_class` — Heuristic combining family size x time span

Hypothesis: Daily -> strongest learning -> calibrated; Episodic -> strongest FLB.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
Recurring	9.1M	+0.0133	+0.0297	+0.0483	+0.0609	+0.0194	+0.0296
		(+0.9)	(+1.2)	(+5.4***)	(+3.8***)	(+1.9)	(+1.7)
Daily	7.4M	-0.0168	-0.0165	+0.0059	-0.0060	-0.0076	-0.0160
		(-2.2)	(-1.9)	(+1.6)	(-0.8)	(-1.8)	(-3.2**)
Episodic	5.0M	+0.0374	+0.0458	+0.0151	+0.0246	+0.0303	+0.0358
		(+6.3***)	(+4.9***)	(+1.7)	(+1.9)	(+3.9***)	(+4.0***)
One-off	2.6M	+0.0476	+0.0567	+0.0456	+0.0523	+0.0428	+0.0493
		(+9.9***)	(+5.9***)	(+8.2***)	(+5.6***)	(+10.8***)	(+6.5***)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

Recurring

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.035	0.033	-0.0021	-0.2	-0.0070	-0.3
D2	0.144	0.119	-0.0244	-1.4	-0.0261	-0.9
D3	0.250	0.349	+0.0991	+1.7	+0.1219	+1.3
D4	0.356	0.198	-0.1586	-1.7	-0.1785	-1.7
D5	0.452	0.317	-0.1353	-1.5	-0.1619	-1.2
D6	0.548	0.731	+0.1831	+1.8	+0.1933	+2.0
D7	0.642	0.835	+0.1927	+1.8	+0.1629	+1.3
D8	0.743	0.617	-0.1254	-1.2	-0.1506	-1.5
D9	0.852	0.896	+0.0444	+2.2	+0.0239	+0.7
D10	0.963	0.975	+0.0112	+1.4	+0.0227	+5.5***
D10-D1			+0.0133	+0.9	+0.0297	+1.2

Daily

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.046	0.059	+0.0124	+2.8*	+0.0140	+2.2
D2	0.145	0.159	+0.0137	+1.8	-0.0000	-0.0
D3	0.246	0.266	+0.0203	+2.6*	-0.0136	-0.9
D4	0.347	0.382	+0.0343	+3.5***	+0.0404	+1.6
D5	0.450	0.495	+0.0450	+5.6***	+0.0417	+3.0*
D6	0.539	0.550	+0.0115	+1.7	+0.0107	+0.8
D7	0.643	0.661	+0.0181	+2.0	+0.0212	+1.2
D8	0.743	0.758	+0.0153	+1.8	+0.0066	+0.3
D9	0.846	0.850	+0.0040	+0.5	-0.0154	-0.9
D10	0.953	0.949	-0.0044	-0.7	-0.0024	-0.4
D10-D1			-0.0168	-2.2	-0.0165	-1.9

Episodic

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.040	0.024	-0.0164	-4.8***	-0.0248	-3.2**
D2	0.147	0.127	-0.0199	-1.9	-0.0179	-0.8
D3	0.245	0.221	-0.0241	-1.4	-0.0300	-0.9
D4	0.345	0.335	-0.0107	-0.5	-0.0244	-0.7
D5	0.445	0.431	-0.0136	-0.6	-0.0131	-0.4
D6	0.544	0.551	+0.0070	+0.4	+0.0056	+0.2
D7	0.645	0.694	+0.0486	+2.1	+0.0439	+1.0
D8	0.747	0.790	+0.0428	+1.9	+0.0496	+1.1
D9	0.848	0.875	+0.0277	+2.0	+0.0242	+0.7
D10	0.962	0.983	+0.0210	+4.3***	+0.0210	+4.1***
D10-D1			+0.0374	+6.3***	+0.0458	+4.9***

11. `dim_group_strict_size` — Family size on event_slug x market_template grouping

Hypothesis: as event_family_size, tighter grouping.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
Singleton	17.8M	+0.0228	+0.0336	+0.0308	+0.0393	+0.0232	+0.0280
1		(+2.7*)	(+1.9)	(+7.6***)	(+5.7***)	(+3.9***)	(+2.6*)
Small	5.0M	+0.0138	+0.0322	+0.0146	+0.0467	+0.0108	+0.0308
2-20		(+1.2)	(+3.3**)	(+1.6)	(+5.7***)	(+1.2)	(+4.3***)

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
Medium 21-1K	1.3M	-0.0041 (-0.3)	+0.0240 (+2.0)	+0.0115 (+0.7)	-0.0596 (-0.9)	+0.0012 (+0.1)	-0.0072 (-0.2)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

Singleton 1

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.037	0.034	-0.0034	-0.4	-0.0104	-0.6
D2	0.146	0.137	-0.0095	-1.0	-0.0223	-1.3
D3	0.248	0.295	+0.0471	+1.9	+0.0355	+0.8
D4	0.352	0.290	-0.0616	-0.9	-0.0940	-1.1
D5	0.448	0.414	-0.0338	-0.7	-0.0880	-0.9
D6	0.545	0.626	+0.0811	+1.2	+0.1151	+1.4
D7	0.642	0.759	+0.1166	+1.5	+0.1067	+1.1
D8	0.743	0.714	-0.0295	-0.7	-0.0416	-0.8
D9	0.848	0.882	+0.0338	+3.4**	+0.0216	+1.1
D10	0.962	0.981	+0.0193	+6.2***	+0.0232	+7.5***
D10-D1			+0.0228	+2.7*	+0.0336	+1.9

Small 2-20

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.042	0.029	-0.0130	-3.8***	-0.0187	-2.5*
D2	0.144	0.130	-0.0143	-1.3	-0.0305	-1.7
D3	0.244	0.226	-0.0181	-1.3	-0.0237	-0.9
D4	0.345	0.340	-0.0053	-0.4	-0.0163	-0.5
D5	0.454	0.467	+0.0128	+1.4	+0.0187	+1.1
D6	0.534	0.563	+0.0292	+3.6***	+0.0248	+1.7
D7	0.645	0.696	+0.0511	+3.0*	+0.0606	+2.6*
D8	0.746	0.775	+0.0296	+2.2	+0.0566	+2.8*
D9	0.849	0.875	+0.0253	+2.4	+0.0413	+2.8*
D10	0.961	0.961	+0.0008	+0.1	+0.0135	+2.2
D10-D1			+0.0138	+1.2	+0.0322	+3.3**

Medium 21-1K

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.043	0.051	+0.0077	+0.8	-0.0053	-0.5
D2	0.141	0.140	-0.0018	-0.1	+0.0677	+0.8
D3	0.243	0.250	+0.0070	+0.1	+0.1359	+0.9
D4	0.347	0.349	+0.0022	+0.1	+0.0953	+1.3
D5	0.447	0.435	-0.0121	-0.6	-0.0438	-1.3
D6	0.539	0.558	+0.0194	+1.0	+0.0234	+0.7
D7	0.645	0.606	-0.0390	-0.9	-0.1418	-1.6
D8	0.748	0.697	-0.0516	-0.6	-0.1779	-0.9
D9	0.849	0.825	-0.0243	-0.5	-0.1535	-1.0
D10	0.957	0.960	+0.0036	+0.4	+0.0187	+2.9*
D10-D1			-0.0041	-0.3	+0.0240	+2.0

12. `dim_event_slug_size` — Family size on event_slug grouping

Hypothesis: as event_family_size at event_slug grain.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
Small	14.0M	+0.0154	+0.0285	+0.0191	+0.0434	+0.0158	+0.0288
2-20		(+1.6)	(+1.2)	(+3.5**)	(+5.8***)	(+2.3)	(+2.2)
Medium	4.9M	+0.0036	+0.0264	+0.0389	+0.0204	+0.0105	+0.0064
21-1K		(+0.2)	(+2.1)	(+5.8***)	(+0.8)	(+1.0)	(+0.3)
Singleton	3.0M	+0.0239	+0.0386	-0.0070	-0.0146	+0.0157	+0.0220
1		(+2.5)	(+3.6***)	(-0.5)	(-0.9)	(+1.9)	(+2.3)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

Small 2-20

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.041	0.038	-0.0026	-0.3	-0.0064	-0.3
D2	0.145	0.142	-0.0039	-0.4	-0.0226	-1.1
D3	0.247	0.288	+0.0406	+1.5	+0.0368	+0.7
D4	0.352	0.269	-0.0834	-1.1	-0.1201	-1.3
D5	0.450	0.400	-0.0499	-0.9	-0.1010	-1.0
D6	0.544	0.645	+0.1011	+1.5	+0.1344	+1.6
D7	0.642	0.780	+0.1378	+1.6	+0.1404	+1.4
D8	0.743	0.708	-0.0357	-0.6	-0.0383	-0.6
D9	0.849	0.882	+0.0329	+3.0**	+0.0280	+1.1
D10	0.960	0.973	+0.0128	+2.1	+0.0221	+6.4***
D10-D1			+0.0154	+1.6	+0.0285	+1.2

Medium 21-1K

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.036	0.038	+0.0025	+0.2	-0.0134	-1.3
D2	0.143	0.124	-0.0196	-1.3	-0.0080	-0.2
D3	0.248	0.269	+0.0208	+0.5	+0.0338	+0.6
D4	0.345	0.372	+0.0270	+0.9	-0.0070	-0.2
D5	0.453	0.487	+0.0342	+2.2	+0.0261	+1.2
D6	0.536	0.525	-0.0108	-0.8	-0.0302	-1.3
D7	0.646	0.656	+0.0099	+0.4	-0.0465	-0.8
D8	0.745	0.736	-0.0086	-0.3	-0.0634	-1.0
D9	0.850	0.862	+0.0122	+0.6	-0.0377	-0.7
D10	0.963	0.969	+0.0061	+0.7	+0.0130	+1.8
D10-D1			+0.0036	+0.2	+0.0264	+2.1

Singleton 1

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.043	0.034	-0.0089	-1.4	-0.0183	-2.2
D2	0.149	0.159	+0.0096	+0.6	+0.0038	+0.1
D3	0.247	0.306	+0.0584	+2.1	+0.0419	+1.1
D4	0.347	0.375	+0.0277	+1.3	+0.0648	+2.0
D5	0.444	0.479	+0.0352	+1.9	+0.0562	+2.7*
D6	0.544	0.549	+0.0052	+0.5	+0.0138	+0.6
D7	0.645	0.648	+0.0031	+0.1	-0.0258	-0.6
D8	0.744	0.724	-0.0199	-0.6	-0.0186	-0.4
D9	0.845	0.855	+0.0093	+0.5	+0.0167	+0.7
D10	0.962	0.977	+0.0150	+2.1	+0.0203	+3.2**

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D10-D1			+0.0239	+2.5	+0.0386	+3.6***

13. `dim_family_vol_tier` — Family total dollar volume terciles

Hypothesis: family dollar volume modulates FLB.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
High vol	23.6M	+0.0187 (+2.7*)	+0.0330 (+2.5)	+0.0266 (+6.6***)	+0.0348 (+4.5***)	+0.0193 (+3.9***)	+0.0275 (+3.3**)
Mid vol	412K	+0.0245 (+5.0***)	+0.0204 (+2.2)	-0.0017 (-0.2)	+0.0020 (+0.3)	+0.0109 (+1.9)	-0.0238 (-2.7*)
Low vol	64K	-0.0116 (-0.6)	+0.0469 (+5.0***)	-0.0004 (-0.1)	+0.0183 (+1.7)	-0.0264 (-1.1)	+0.0035 (+0.2)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

High vol

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.034	-0.0047	-0.8	-0.0120	-0.9
D2	0.145	0.135	-0.0100	-1.4	-0.0193	-1.4
D3	0.247	0.279	+0.0320	+1.7	+0.0297	+0.8
D4	0.351	0.298	-0.0525	-0.9	-0.0843	-1.1
D5	0.449	0.424	-0.0255	-0.6	-0.0717	-0.8
D6	0.543	0.613	+0.0698	+1.3	+0.0996	+1.3
D7	0.643	0.748	+0.1055	+1.5	+0.0997	+1.2
D8	0.744	0.724	-0.0204	-0.6	-0.0315	-0.7
D9	0.849	0.877	+0.0283	+3.3**	+0.0184	+1.0
D10	0.961	0.975	+0.0140	+3.7***	+0.0210	+7.2***
D10-D1			+0.0187	+2.7*	+0.0330	+2.5

Mid vol

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.044	0.042	-0.0020	-0.5	+0.0029	+0.3
D2	0.144	0.142	-0.0024	-0.3	+0.0077	+0.5
D3	0.244	0.231	-0.0131	-0.8	-0.0050	-0.3
D4	0.347	0.346	-0.0009	-0.1	+0.0154	+0.8
D5	0.446	0.491	+0.0452	+3.5**	+0.0622	+2.2
D6	0.540	0.565	+0.0255	+2.3	-0.0030	-0.2
D7	0.642	0.683	+0.0405	+3.2**	+0.0598	+3.2**
D8	0.745	0.760	+0.0151	+1.3	+0.0126	+0.8
D9	0.845	0.878	+0.0334	+4.3***	+0.0286	+2.6*
D10	0.952	0.974	+0.0225	+9.2***	+0.0233	+6.8***
D10-D1			+0.0245	+5.0***	+0.0204	+2.2

Low vol

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.045	0.050	+0.0056	+0.4	-0.0297	-4.8***
D2	0.142	0.136	-0.0059	-0.5	-0.0343	-1.9

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D3	0.245	0.266	+0.0210	+0.9	+0.0258	+0.9
D4	0.345	0.357	+0.0126	+0.6	+0.0127	+0.4
D5	0.445	0.485	+0.0404	+1.3	+0.0334	+1.1
D6	0.537	0.579	+0.0420	+2.1	+0.0410	+1.7
D7	0.640	0.704	+0.0642	+2.2	+0.0498	+1.3
D8	0.743	0.771	+0.0276	+1.3	+0.0207	+0.8
D9	0.846	0.893	+0.0465	+4.0***	+0.0681	+5.3***
D10	0.951	0.945	-0.0061	-0.4	+0.0172	+2.5
D10-D1			-0.0116	-0.6	+0.0469	+5.0***

14. `dim_family_size_x_vol` — 3x3 cross-tab of family size x volume

Hypothesis: disentangle family count vs family dollar attention.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
Small 2-20 × High vol	9.0M	+0.0451 (+11.2***)	+0.0583 (+8.1***)	+0.0339 (+3.8***)	+0.0504 (+3.8***)	+0.0395 (+7.1***)	+0.0521 (+7.5***)
Large 1K+ × High vol	6.6M	-0.0193 (-2.1)	-0.0152 (-1.6)	+0.0064 (+1.6)	-0.0053 (-0.7)	-0.0081 (-1.7)	-0.0132 (-2.5)
Medium 21-1K × High vol	5.6M	-0.0025 (-0.1)	-0.0010 (-0.0)	+0.0332 (+3.8***)	+0.0398 (+2.3)	+0.0061 (+0.5)	+0.0030 (+0.1)
Singleton 1 × High vol	2.4M	+0.0485 (+9.4***)	+0.0572 (+5.8***)	+0.0478 (+8.0***)	+0.0527 (+5.4***)	+0.0435 (+10.0***)	+0.0499 (+6.3***)
Small 2-20 × Mid vol	232K	+0.0040 (+0.5)	-0.0042 (-0.2)	-0.0177 (-1.2)	-0.0240 (-2.0)	-0.0192 (-1.8)	-0.0930 (-5.9***)
Singleton 1 × Mid vol	173K	+0.0394 (+6.2***)	+0.0360 (+3.2**)	+0.0252 (+4.0***)	+0.0351 (+4.4***)	+0.0397 (+9.2***)	+0.0324 (+4.1***)
Singleton 1 × Low vol	34K	+0.0037 (+0.2)	+0.0527 (+5.2***)	+0.0201 (+2.5*)	+0.0471 (+4.7***)	+0.0193 (+2.0)	+0.0405 (+2.3)
Small 2-20 × Low vol	29K	-0.0474 (-1.3)	+0.0277 (+1.3)	-0.0232 (-1.9)	-0.0193 (-0.9)	-0.0921 (-1.8)	-0.0652 (-2.1)
Medium 21-1K × Mid vol	8K	+0.0002 (+0.0)	+0.0353 (+1.4)	+0.0057 (+0.3)	+0.0176 (+0.5)	-0.0006 (-0.1)	+0.0312 (+1.8)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

Small 2-20 × High vol

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.040	0.021	-0.0198	-6.9***	-0.0330	-5.5***
D2	0.146	0.134	-0.0120	-0.8	-0.0183	-0.7
D3	0.247	0.312	+0.0642	+1.3	+0.0618	+0.9
D4	0.355	0.202	-0.1535	-1.8	-0.1725	-1.8
D5	0.451	0.321	-0.1294	-1.6	-0.1521	-1.2
D6	0.548	0.714	+0.1662	+1.7	+0.1852	+2.0
D7	0.642	0.824	+0.1822	+1.9	+0.1650	+1.5

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D8	0.744	0.662	-0.0823	-0.9	-0.0737	-0.8
D9	0.850	0.888	+0.0381	+2.2	+0.0236	+0.8
D10	0.962	0.987	+0.0254	+9.0***	+0.0253	+6.4***
D10-D1			+0.0451	+11.2***	+0.0583	+8.1***

Large 1K+ × High vol

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.047	0.060	+0.0129	+2.6*	+0.0131	+2.0
D2	0.145	0.158	+0.0126	+1.4	+0.0005	+0.0
D3	0.246	0.267	+0.0201	+2.3	-0.0136	-0.8
D4	0.348	0.381	+0.0331	+3.2**	+0.0299	+1.2
D5	0.450	0.493	+0.0433	+5.1***	+0.0398	+3.0*
D6	0.538	0.551	+0.0129	+1.8	+0.0158	+1.2
D7	0.642	0.661	+0.0184	+1.9	+0.0221	+1.2
D8	0.743	0.761	+0.0183	+2.0	+0.0094	+0.3
D9	0.846	0.850	+0.0046	+0.5	-0.0164	-0.9
D10	0.954	0.947	-0.0064	-0.8	-0.0021	-0.3
D10-D1			-0.0193	-2.1	-0.0152	-1.6

Medium 21-1K × High vol

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.034	0.039	+0.0047	+0.3	+0.0163	+0.5
D2	0.144	0.114	-0.0302	-2.3	-0.0306	-1.1
D3	0.248	0.261	+0.0138	+0.3	+0.0270	+0.6
D4	0.344	0.391	+0.0461	+1.7	+0.0808	+2.1
D5	0.447	0.478	+0.0313	+1.5	+0.0146	+0.4
D6	0.541	0.536	-0.0054	-0.3	-0.0464	-1.4
D7	0.646	0.661	+0.0148	+0.6	-0.0657	-1.2
D8	0.746	0.761	+0.0144	+0.5	-0.0233	-0.4
D9	0.850	0.879	+0.0289	+1.7	+0.0219	+0.6
D10	0.963	0.965	+0.0022	+0.2	+0.0154	+3.0**
D10-D1			-0.0025	-0.1	-0.0010	-0.0

15. `dim_vol_per_contract_tier` — Dollar-per-contract quintiles

Hypothesis: higher per-contract dollar attention -> calibrated.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
VPC Q5	20.3M	+0.0228	+0.0349	+0.0308	+0.0388	+0.0238	+0.0315
(thickest)		(+3.0**)	(+2.5*)	(+6.4***)	(+4.7***)	(+4.3***)	(+3.6***)
VPC Q4	2.7M	-0.0172	-0.0118	+0.0097	-0.0067	-0.0065	-0.0217
		(-2.0)	(-1.0)	(+2.9*)	(-0.8)	(-1.3)	(-3.4**)
VPC Q3	858K	+0.0090	+0.0052	+0.0133	+0.0097	+0.0067	-0.0429
		(+2.0)	(+0.7)	(+3.3**)	(+1.4)	(+2.0)	(-4.7***)
VPC Q2	166K	+0.0005	-0.0635	-0.0181	-0.0776	-0.0227	-0.1803
		(+0.1)	(-1.4)	(-0.9)	(-2.8*)	(-2.1)	(-4.8***)
VPC Q1	22K	-0.1442	-0.0375	-0.0376	-0.0410	-0.1072	-0.1078
(thinnest)		(-2.0)	(-1.1)	(-2.0)	(-1.2)	(-2.2)	(-2.1)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

VPC Q5 (thickest)

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.038	0.031	-0.0071	-1.1	-0.0131	-1.0
D2	0.145	0.127	-0.0186	-2.2	-0.0221	-1.5
D3	0.248	0.279	+0.0309	+1.3	+0.0294	+0.8
D4	0.351	0.282	-0.0695	-1.1	-0.0903	-1.1
D5	0.450	0.410	-0.0402	-0.8	-0.0791	-0.9
D6	0.544	0.619	+0.0755	+1.2	+0.1049	+1.3
D7	0.643	0.757	+0.1142	+1.4	+0.1031	+1.1
D8	0.744	0.712	-0.0315	-0.7	-0.0369	-0.8
D9	0.849	0.878	+0.0294	+2.9*	+0.0177	+0.9
D10	0.962	0.978	+0.0157	+3.9***	+0.0218	+7.3***
D10-D1			+0.0228	+3.0**	+0.0349	+2.5*

VPC Q4

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.043	0.056	+0.0134	+3.4**	+0.0132	+1.3
D2	0.146	0.176	+0.0298	+4.0***	+0.0205	+1.5
D3	0.245	0.277	+0.0321	+4.6***	+0.0283	+1.6
D4	0.346	0.385	+0.0394	+5.4***	+0.0232	+1.5
D5	0.448	0.503	+0.0554	+7.3***	+0.0310	+1.8
D6	0.539	0.577	+0.0379	+6.2***	+0.0379	+1.6
D7	0.643	0.692	+0.0495	+7.1***	+0.0431	+2.0
D8	0.744	0.779	+0.0356	+5.4***	+0.0410	+2.0
D9	0.846	0.868	+0.0223	+3.5***	+0.0315	+3.0*
D10	0.954	0.950	-0.0038	-0.5	+0.0014	+0.2
D10-D1			-0.0172	-2.0	-0.0118	-1.0

VPC Q3

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.045	0.051	+0.0064	+1.8	+0.0068	+1.1
D2	0.144	0.159	+0.0151	+2.0	+0.0069	+0.6
D3	0.243	0.278	+0.0349	+2.5	+0.0390	+2.0
D4	0.344	0.396	+0.0526	+5.7***	-0.0045	-0.2
D5	0.446	0.495	+0.0495	+5.3***	+0.0810	+1.3
D6	0.540	0.585	+0.0450	+5.1***	-0.0675	-1.6
D7	0.644	0.673	+0.0297	+3.6***	+0.0248	+1.2
D8	0.745	0.769	+0.0245	+2.9*	+0.0051	+0.3
D9	0.848	0.880	+0.0323	+5.3***	+0.0191	+1.4
D10	0.953	0.968	+0.0154	+5.7***	+0.0120	+2.9*
D10-D1			+0.0090	+2.0	+0.0052	+0.7

16. `dim_vol_per_contract_residualized` — Volume per contract, residualized on size

Hypothesis: volume per contract conditional on size.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
VPC resid	22.5M	+0.0203	+0.0337	+0.0279	+0.0357	+0.0211	+0.0293
High		(+2.8*)	(+2.5*)	(+6.6***)	(+4.6***)	(+4.1***)	(+3.5**)

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
VPC resid	1.5M	-0.0020	-0.0180	+0.0068	-0.0082	-0.0019	-0.0705
Mid		(-0.5)	(-2.0)	(+1.6)	(-1.2)	(-0.6)	(-6.9***)
VPC resid	93K	-0.0545	-0.0572	-0.0092	-0.0468	-0.0552	-0.1190
Low		(-2.3)	(-1.6)	(-1.1)	(-1.7)	(-2.1)	(-5.2***)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

VPC resid High

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.033	-0.0061	-1.0	-0.0125	-1.0
D2	0.145	0.132	-0.0131	-1.7	-0.0200	-1.4
D3	0.247	0.278	+0.0303	+1.5	+0.0292	+0.8
D4	0.351	0.293	-0.0581	-1.0	-0.0850	-1.1
D5	0.450	0.421	-0.0286	-0.7	-0.0730	-0.9
D6	0.543	0.614	+0.0709	+1.3	+0.1007	+1.3
D7	0.643	0.751	+0.1081	+1.5	+0.1000	+1.1
D8	0.744	0.721	-0.0228	-0.6	-0.0319	-0.7
D9	0.849	0.878	+0.0289	+3.2**	+0.0185	+1.0
D10	0.962	0.976	+0.0142	+3.7***	+0.0212	+7.3***
D10-D1			+0.0203	+2.8*	+0.0337	+2.5*

VPC resid Mid

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.044	0.059	+0.0148	+4.4***	+0.0265	+3.3**
D2	0.144	0.164	+0.0201	+3.3**	+0.0188	+1.8
D3	0.244	0.281	+0.0374	+3.5***	+0.0429	+2.7*
D4	0.344	0.394	+0.0500	+6.9***	+0.0091	+0.5
D5	0.445	0.496	+0.0501	+7.0***	+0.0772	+2.0
D6	0.540	0.578	+0.0388	+5.2***	-0.0306	-1.1
D7	0.643	0.676	+0.0328	+5.0***	+0.0595	+2.6*
D8	0.745	0.769	+0.0241	+3.9***	+0.0151	+1.3
D9	0.847	0.871	+0.0245	+4.1***	+0.0184	+1.7
D10	0.953	0.966	+0.0128	+5.9***	+0.0085	+2.3
D10-D1			-0.0020	-0.5	-0.0180	-2.0

VPC resid Low

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.045	0.075	+0.0301	+2.0	+0.0256	+0.9
D2	0.142	0.178	+0.0359	+3.0*	+0.0206	+1.0
D3	0.244	0.284	+0.0406	+2.1	+0.0382	+1.4
D4	0.348	0.327	-0.0213	-0.6	-0.0079	-0.3
D5	0.444	0.471	+0.0273	+1.0	+0.0031	+0.1
D6	0.536	0.538	+0.0018	+0.1	+0.0218	+0.7
D7	0.638	0.705	+0.0673	+1.8	+0.1085	+1.8
D8	0.744	0.745	+0.0001	+0.0	-0.0073	-0.3
D9	0.844	0.842	-0.0018	-0.1	-0.0098	-0.3
D10	0.949	0.925	-0.0244	-1.3	-0.0316	-1.4
D10-D1			-0.0545	-2.3	-0.0572	-1.6

17. `dim_text_novelty` — Semantic isolation by best-neighbor cosine (fixed thresholds)

Hypothesis: slugs with no near-duplicate (semantic novelty) -> no prior anchor -> strong FLB.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
>0.95 near duplicate	18.5M	+0.0099 (+1.0)	+0.0272 (+1.6)	+0.0210 (+4.4***)	+0.0361 (+5.2***)	+0.0144 (+2.2)	+0.0244 (+2.3)
0.90-0.95 close lex match	2.4M	+0.0135 (+1.5)	+0.0251 (+2.1)	+0.0173 (+1.8)	+0.0268 (+2.3)	+0.0072 (+0.9)	+0.0090 (+0.7)
<0.50 genuinely isolated	2.2M	+0.0556 (+11.8***)	+0.0684 (+6.7***)	+0.0605 (+14.2***)	+0.0690 (+8.4***)	+0.0518 (+12.8***)	+0.0654 (+8.9***)
0.75-0.90 has neighbor	954K	+0.0509 (+5.2***)	+0.0722 (+16.9***)	+0.0101 (+0.5)	-0.0584 (-0.9)	+0.0314 (+3.5**)	+0.0236 (+0.9)
0.50-0.75 mod isolated	23K	+0.0599 (+7.9***)	+0.0796 (+8.3***)	+0.0757 (+9.2***)	+0.0913 (+13.5***)	+0.0634 (+9.9***)	+0.0895 (+14.1***)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

>0.95 near duplicate

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.038	0.039	+0.0008	+0.1	-0.0074	-0.4
D2	0.146	0.140	-0.0061	-0.8	-0.0291	-1.9
D3	0.248	0.297	+0.0494	+2.1	+0.0367	+0.9
D4	0.351	0.292	-0.0587	-0.9	-0.0945	-1.1
D5	0.450	0.422	-0.0283	-0.6	-0.0754	-0.9
D6	0.543	0.619	+0.0759	+1.3	+0.1054	+1.4
D7	0.643	0.760	+0.1170	+1.5	+0.1135	+1.2
D8	0.743	0.711	-0.0318	-0.7	-0.0306	-0.6
D9	0.848	0.875	+0.0264	+2.7*	+0.0226	+1.2
D10	0.961	0.972	+0.0107	+2.1	+0.0198	+5.6***
D10-D1			+0.0099	+1.0	+0.0272	+1.6

0.90-0.95 close lex match

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.043	0.040	-0.0025	-0.4	-0.0124	-1.3
D2	0.142	0.140	-0.0025	-0.1	+0.0396	+1.1
D3	0.243	0.220	-0.0233	-1.1	+0.0125	+0.3
D4	0.345	0.317	-0.0277	-1.4	-0.0190	-0.7
D5	0.446	0.415	-0.0309	-1.1	-0.0215	-0.8
D6	0.543	0.542	-0.0006	-0.0	-0.0510	-1.6
D7	0.646	0.628	-0.0183	-0.5	-0.1133	-1.3
D8	0.747	0.744	-0.0032	-0.1	-0.0878	-1.1
D9	0.850	0.870	+0.0201	+1.2	-0.0016	-0.1
D10	0.958	0.970	+0.0110	+1.8	+0.0127	+1.7
D10-D1			+0.0135	+1.5	+0.0251	+2.1

<0.50 genuinely isolated

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.012	-0.0271	-8.4***	-0.0371	-4.5***
D2	0.144	0.099	-0.0449	-1.9	-0.0639	-3.2**
D3	0.244	0.175	-0.0687	-2.4	-0.0771	-2.4
D4	0.349	0.316	-0.0335	-0.9	-0.0387	-0.9
D5	0.445	0.438	-0.0070	-0.3	-0.0208	-0.7
D6	0.545	0.574	+0.0296	+1.0	+0.0101	+0.3
D7	0.644	0.738	+0.0940	+3.5***	+0.0898	+2.6*
D8	0.746	0.828	+0.0817	+3.3**	+0.0778	+2.6*
D9	0.850	0.914	+0.0638	+4.3***	+0.0676	+3.3**
D10	0.962	0.991	+0.0285	+8.3***	+0.0312	+5.3***
D10-D1			+0.0556	+11.8***	+0.0684	+6.7***

18. `dim_text_neighbors_strict` — Count of slugs above 0.75 cosine

Hypothesis: as text_novelty, operationalized by threshold-count.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
6+ strict neighbors	21.8M	+0.0127 (+1.6)	+0.0299 (+2.1)	+0.0203 (+4.5***)	+0.0288 (+3.3**)	+0.0144 (+2.6*)	+0.0230 (+2.5*)
0 strict neighbors	2.3M	+0.0556 (+11.9***)	+0.0684 (+6.8***)	+0.0606 (+14.3***)	+0.0691 (+8.5***)	+0.0519 (+13.0***)	+0.0656 (+9.0***)
2-5 strict neighbors	18K	+0.0391 (+2.1)	+0.0354 (+1.4)	-0.1152 (-0.9)	-0.1678 (-0.8)	+0.0017 (+0.1)	-0.0163 (-0.2)
1 strict neighbor	—	—	—	—	—	+0.0375 (+1.7)	+0.0737 (+5.4***)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

6+ strict neighbors

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.038	-0.0012	-0.2	-0.0100	-0.7
D2	0.145	0.139	-0.0066	-0.9	-0.0160	-1.1
D3	0.247	0.286	+0.0391	+1.9	+0.0368	+1.0
D4	0.351	0.298	-0.0527	-0.9	-0.0863	-1.1
D5	0.450	0.425	-0.0251	-0.6	-0.0727	-0.8
D6	0.543	0.614	+0.0711	+1.3	+0.1015	+1.3
D7	0.643	0.748	+0.1050	+1.4	+0.1004	+1.1
D8	0.744	0.716	-0.0280	-0.7	-0.0380	-0.9
D9	0.848	0.873	+0.0250	+2.8*	+0.0148	+0.8
D10	0.961	0.973	+0.0115	+2.6*	+0.0199	+6.4***
D10-D1			+0.0127	+1.6	+0.0299	+2.1

0 strict neighbors

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.012	-0.0271	-8.4***	-0.0372	-4.5***
D2	0.144	0.100	-0.0445	-1.9	-0.0635	-3.2**
D3	0.244	0.176	-0.0684	-2.4	-0.0747	-2.4
D4	0.349	0.314	-0.0349	-0.9	-0.0380	-0.9
D5	0.445	0.435	-0.0092	-0.4	-0.0184	-0.6
D6	0.545	0.575	+0.0302	+1.0	+0.0095	+0.3

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D7	0.644	0.740	+0.0955	+3.6***	+0.0893	+2.6*
D8	0.746	0.828	+0.0819	+3.3**	+0.0776	+2.7*
D9	0.850	0.912	+0.0620	+4.2***	+0.0666	+3.3**
D10	0.962	0.991	+0.0285	+8.4***	+0.0313	+5.4***
D10-D1			+0.0556	+11.9***	+0.0684	+6.8***

2-5 strict neighbors

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.043	0.032	-0.0108	-0.7	-0.0266	-1.8
D2	0.148	0.216	+0.0679	+0.8	+0.3775	+1.8
D3	0.242	0.336	+0.0933	+0.6	+0.3154	+1.8
D4	0.345	0.506	+0.1609	+1.2	+0.4834	+6.2***
D5	0.439	0.537	+0.0980	+0.7	+0.2860	+1.5
D6	0.547	0.519	-0.0281	-0.4	-0.0636	-0.4
D7	0.646	0.665	+0.0197	+0.2	-0.3987	-3.7***
D8	0.748	0.660	-0.0876	-0.6	-0.3484	-1.9
D9	0.847	0.778	-0.0689	-0.7	-0.2902	-1.2
D10	0.956	0.985	+0.0283	+3.0**	+0.0088	+0.4
D10-D1			+0.0391	+2.1	+0.0354	+1.4

19. `dim_prior_settlements_bin__event_template` — Prior settled contracts at trade time

Hypothesis: more prior same-template settlements at trade time -> less FLB.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
0	13.1M	+0.0319	+0.0376	+0.0462	+0.0643	+0.0322	+0.0404
		(+4.3***)	(+2.0)	(+11.8***)	(+10.3***)	(+5.8***)	(+3.5***)
50+	7.6M	-0.0273	-0.0083	+0.0003	-0.0091	-0.0125	-0.0114
		(-1.9)	(-1.0)	(+0.0)	(-1.2)	(-1.5)	(-2.1)
6-50	1.8M	+0.0133	+0.0337	+0.0509	+0.0696	+0.0170	+0.0190
		(+0.5)	(+2.0)	(+10.6***)	(+12.9***)	(+0.9)	(+0.7)
1-5	1.5M	+0.0424	+0.0455	-0.0051	-0.0394	+0.0261	+0.0142
		(+3.7***)	(+2.2)	(-0.2)	(-0.8)	(+2.0)	(+0.6)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

0

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.038	0.027	-0.0103	-1.5	-0.0125	-0.7
D2	0.145	0.125	-0.0201	-1.7	-0.0282	-1.5
D3	0.247	0.290	+0.0427	+1.2	+0.0426	+0.7
D4	0.354	0.231	-0.1226	-1.6	-0.1453	-1.6
D5	0.450	0.347	-0.1028	-1.4	-0.1410	-1.2
D6	0.547	0.682	+0.1345	+1.5	+0.1630	+1.7
D7	0.642	0.800	+0.1576	+1.7	+0.1389	+1.3
D8	0.744	0.686	-0.0581	-0.9	-0.0488	-0.7
D9	0.850	0.895	+0.0453	+3.9***	+0.0328	+1.4
D10	0.963	0.984	+0.0215	+7.7***	+0.0252	+8.2***
D10-D1			+0.0319	+4.3***	+0.0376	+2.0

50+

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.046	0.057	+0.0113	+2.6*	+0.0095	+1.5
D2	0.145	0.157	+0.0121	+1.5	-0.0011	-0.1
D3	0.246	0.264	+0.0180	+2.2	-0.0114	-0.8
D4	0.347	0.380	+0.0329	+3.3**	+0.0388	+1.6
D5	0.450	0.494	+0.0448	+5.6***	+0.0421	+3.0**
D6	0.539	0.549	+0.0108	+1.6	+0.0099	+0.8
D7	0.643	0.664	+0.0211	+2.3	+0.0211	+1.2
D8	0.743	0.759	+0.0165	+1.9	+0.0036	+0.1
D9	0.846	0.847	+0.0011	+0.1	-0.0156	-0.9
D10	0.955	0.939	-0.0160	-1.1	+0.0012	+0.2
D10-D1			-0.0273	-1.9	-0.0083	-1.0

6-50

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.033	0.037	+0.0040	+0.2	-0.0181	-1.4
D2	0.143	0.089	-0.0544	-4.1***	-0.0534	-3.1**
D3	0.253	0.301	+0.0481	+0.5	+0.0241	+0.4
D4	0.343	0.416	+0.0727	+1.2	+0.0956	+1.4
D5	0.445	0.453	+0.0084	+0.3	-0.0298	-0.6
D6	0.542	0.577	+0.0353	+1.2	+0.0502	+0.8
D7	0.648	0.669	+0.0211	+0.5	-0.0073	-0.1
D8	0.744	0.739	-0.0049	-0.1	-0.0418	-0.5
D9	0.852	0.906	+0.0540	+3.5***	+0.0644	+2.7*
D10	0.962	0.979	+0.0174	+1.7	+0.0156	+1.6
D10-D1			+0.0133	+0.5	+0.0337	+2.0

20. `dim_prior_settlements_bin__event_slug` — Prior settled contracts (event_slug grouping)

Hypothesis: same at event_slug grain.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
0	23.1M	+0.0175 (+2.5*)	+0.0313 (+2.3)	+0.0271 (+6.9***)	+0.0411 (+7.4***)	+0.0185 (+3.7***)	+0.0275 (+3.2**)
1-5	783K	+0.0342 (+5.5***)	+0.0584 (+10.0***)	-0.0184 (-0.8)	-0.1356 (-1.3)	+0.0209 (+2.6*)	-0.0015 (-0.0)
6-50	246K	+0.0628 (+11.7***)	+0.0837 (+19.2***)	+0.0371 (+3.6***)	+0.0403 (+3.2**)	+0.0422 (+5.1***)	+0.0558 (+8.3***)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

0

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.035	-0.0041	-0.7	-0.0108	-0.8
D2	0.145	0.135	-0.0106	-1.5	-0.0230	-1.7
D3	0.247	0.278	+0.0305	+1.6	+0.0238	+0.7
D4	0.351	0.297	-0.0535	-0.9	-0.0863	-1.1
D5	0.449	0.424	-0.0248	-0.6	-0.0723	-0.8
D6	0.543	0.614	+0.0711	+1.3	+0.1009	+1.3
D7	0.643	0.750	+0.1071	+1.5	+0.1030	+1.2

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D8	0.744	0.725	-0.0188	-0.5	-0.0271	-0.6
D9	0.849	0.879	+0.0309	+3.8***	+0.0245	+1.5
D10	0.961	0.975	+0.0134	+3.4**	+0.0205	+6.6***
D10-D1			+0.0175	+2.5*	+0.0313	+2.3

1-5

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.028	-0.0117	-2.1	-0.0348	-7.5***
D2	0.146	0.165	+0.0192	+0.4	+0.1038	+0.7
D3	0.246	0.282	+0.0361	+0.4	+0.1562	+1.0
D4	0.345	0.347	+0.0020	+0.1	+0.0360	+0.5
D5	0.453	0.440	-0.0129	-0.8	-0.0323	-1.5
D6	0.534	0.561	+0.0272	+1.7	+0.0515	+2.1
D7	0.646	0.677	+0.0312	+0.7	-0.0726	-1.0
D8	0.747	0.727	-0.0200	-0.2	-0.1106	-0.6
D9	0.847	0.828	-0.0182	-0.3	-0.1238	-0.8
D10	0.963	0.985	+0.0226	+8.5***	+0.0236	+6.6***
D10-D1			+0.0342	+5.5***	+0.0584	+10.0***

6-50

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.041	0.009	-0.0324	-10.3***	-0.0470	-13.9***
D2	0.141	0.118	-0.0226	-0.6	-0.0422	-0.9
D3	0.247	0.306	+0.0587	+1.1	+0.1002	+1.0
D4	0.343	0.384	+0.0407	+0.7	+0.0667	+0.5
D5	0.448	0.458	+0.0097	+0.2	-0.0069	-0.1
D6	0.536	0.503	-0.0329	-1.1	-0.0582	-1.5
D7	0.647	0.609	-0.0376	-0.6	-0.0855	-0.8
D8	0.746	0.656	-0.0894	-1.3	-0.0993	-1.1
D9	0.854	0.863	+0.0086	+0.2	+0.0397	+0.7
D10	0.959	0.989	+0.0304	+7.0***	+0.0367	+13.4***
D10-D1			+0.0628	+11.7***	+0.0837	+19.2***

21. `dim_prior_settlements_bin__dim_group_strict` — Prior settled contracts (strict grouping)

Hypothesis: same at strict grouping grain.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
0	23.3M	+0.0172 (+2.5)	+0.0309 (+2.2)	+0.0266 (+6.9***)	+0.0406 (+7.4***)	+0.0182 (+3.7***)	+0.0272 (+3.2**)
1-5	520K	+0.0623 (+8.7***)	+0.0825 (+20.0***)	-0.0328 (-0.8)	-0.1611 (-1.3)	+0.0343 (+2.7*)	+0.0007 (+0.0)
6-50	228K	+0.0628 (+10.2***)	+0.0802 (+15.7***)	+0.0454 (+4.7***)	+0.0433 (+3.5***)	+0.0494 (+5.5***)	+0.0616 (+10.9***)
50+	31K	+0.1080 (+292.3***)	+0.1109 (+0.0)	+0.0566 (+0.0)	+0.0683 (+339.3***)	+0.0620 (+1504.2***)	+0.0726 (+848.0***)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

0

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.035	-0.0039	-0.7	-0.0107	-0.8
D2	0.145	0.135	-0.0099	-1.4	-0.0224	-1.7
D3	0.247	0.278	+0.0307	+1.6	+0.0244	+0.7
D4	0.351	0.298	-0.0530	-0.9	-0.0861	-1.1
D5	0.449	0.425	-0.0243	-0.6	-0.0713	-0.8
D6	0.543	0.613	+0.0702	+1.3	+0.0997	+1.3
D7	0.643	0.749	+0.1066	+1.5	+0.1020	+1.2
D8	0.744	0.725	-0.0186	-0.5	-0.0265	-0.6
D9	0.849	0.878	+0.0299	+3.7***	+0.0243	+1.5
D10	0.961	0.975	+0.0133	+3.4**	+0.0203	+6.6***
D10-D1			+0.0172	+2.5	+0.0309	+2.2

1-5

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.043	0.015	-0.0284	-4.1***	-0.0493	-18.2***
D2	0.147	0.141	-0.0065	-0.1	+0.0793	+0.5
D3	0.246	0.287	+0.0410	+0.4	+0.1493	+0.8
D4	0.343	0.333	-0.0097	-0.2	+0.0382	+0.4
D5	0.449	0.432	-0.0167	-0.9	-0.0493	-1.2
D6	0.538	0.568	+0.0298	+1.4	+0.0657	+2.0
D7	0.649	0.679	+0.0299	+0.4	-0.0804	-0.8
D8	0.748	0.712	-0.0361	-0.3	-0.1468	-0.7
D9	0.847	0.845	-0.0017	-0.0	-0.1180	-0.7
D10	0.959	0.993	+0.0339	+16.8***	+0.0332	+10.7***
D10-D1			+0.0623	+8.7***	+0.0825	+20.0***

6-50

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.040	0.008	-0.0319	-9.0***	-0.0437	-10.8***
D2	0.142	0.139	-0.0031	-0.1	-0.0139	-0.2
D3	0.246	0.305	+0.0592	+1.1	+0.1850	+1.5
D4	0.347	0.410	+0.0638	+1.2	+0.2552	+2.4
D5	0.450	0.425	-0.0244	-0.7	-0.0996	-1.1
D6	0.535	0.564	+0.0295	+1.0	+0.0445	+0.5
D7	0.645	0.598	-0.0466	-0.7	-0.1488	-1.2
D8	0.745	0.651	-0.0934	-1.3	-0.1133	-0.9
D9	0.853	0.848	-0.0050	-0.1	+0.0160	+0.2
D10	0.959	0.990	+0.0309	+6.2***	+0.0366	+11.8***
D10-D1			+0.0628	+10.2***	+0.0802	+15.7***

22. `dim_market_type` — Up/down vs non-up/down sensitivity

Hypothesis: up/down crypto markets are noise-trading regimes structurally different from learnability markets.

Spread by window x weighting (cnt = count-weighted, $ = dollar-weighted; each cell `spread (t)`):

Slice	N	25-80% cnt	25-80% $	80-100% cnt	80-100% $	full cnt	full $
non_updown	24.0M	+0.0187 (+2.8*)	+0.0330 (+2.5)	+0.0255 (+6.5***)	+0.0345 (+4.6***)	+0.0189 (+3.9***)	+0.0270 (+3.3**)
updown	69K	+0.0519 (+55.5***)	+0.0463 (+56.3***)	+0.0548 (+350.4***)	+0.0670 (+1684.3***)	+0.0545 (+922.0***)	+0.0653 (+0.0)

Decile breakdown (mature 25-80%, top 3 slices by trade count):

non_updown

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.039	0.034	-0.0046	-0.8	-0.0119	-0.9
D2	0.145	0.135	-0.0099	-1.4	-0.0191	-1.4
D3	0.247	0.278	+0.0314	+1.7	+0.0294	+0.8
D4	0.351	0.299	-0.0515	-0.9	-0.0838	-1.1
D5	0.449	0.425	-0.0242	-0.6	-0.0712	-0.8
D6	0.543	0.612	+0.0694	+1.3	+0.0993	+1.3
D7	0.643	0.747	+0.1045	+1.5	+0.0996	+1.2
D8	0.744	0.724	-0.0198	-0.6	-0.0312	-0.7
D9	0.849	0.877	+0.0285	+3.4**	+0.0184	+1.0
D10	0.961	0.975	+0.0141	+3.8***	+0.0210	+7.3***
D10-D1			+0.0187	+2.8*	+0.0330	+2.5

updown

Decile	Impl. prob	Win rate	Return (cnt)	t cnt	Return ($)	t $
D1	0.066	0.055	-0.0111	-11.9***	-0.0021	-2.5*
D2	0.144	0.143	-0.0016	-1.1	+0.0351	+6.0***
D3	0.245	0.181	-0.0645	-129.9***	-0.0986	-171.7***
D4	0.346	0.296	-0.0498	-15.4***	-0.1646	-10.4***
D5	0.453	0.435	-0.0175	-4.4***	-0.0401	-3.0*
D6	0.531	0.545	+0.0140	+18.0***	+0.0096	+1.5
D7	0.644	0.714	+0.0697	+24.3***	+0.0263	+4.5***
D8	0.739	0.799	+0.0605	+66.7***	+0.0102	+0.9
D9	0.837	0.857	+0.0204	+0.0	+0.0467	+0.0
D10	0.942	0.983	+0.0408	+1966.0***	+0.0442	+17257.2***
D10-D1			+0.0519	+55.5***	+0.0463	+56.3***

Appendix A — Jaccard overlap between headline contract sets

The audit (Phase 1.1) tested whether the "convergent" headline findings are one underlying contract set. Set sizes: v3 Small 2-20 = 150,205; Small×HighVol = 55,651; Q1 text-isolated = 135,302; 0-strict-neighbors = 17,761; 0-priors event_template = 148,174.

	v3_Small	Small×HiVol	Q1_iso	0_neigh	0_priors
v3 Small 2-20	1.00	0.37	0.04	0.05	0.56
Small × HighVol	0.37	1.00	0.03	0.07	0.29
Q1 isolated	0.04	0.03	1.00	0.13	0.04
0 neighbors	0.05	0.07	0.13	1.00	0.05
0 priors	0.56	0.29	0.04	0.05	1.00

Max off-diagonal 0.56 (Small 2-20 ↔ 0-priors). The text-novelty cuts overlap only 0.13 with each other and <=0.07 with the family-size cuts. The headline findings are not one contract set — each identifies an independent sub-population (though composition-concentrated: Q1 isolated is 90% Crypto / 86.7% up/down before exclusion; Small×HighVol is 52% Sports).

Appendix B — Up/down sensitivity

Up/down markets are 41.1% of contracts and were excluded from the primary v6 view; `dim_market_type` reports them. The audit's up/down-included-vs-excluded re-run found three patterns: (1) low-up/down headline cells unchanged (Small 2-20 × High vol +0.0501 either way; event_slug 50+ priors +0.0805 either way; 0-priors event_template +0.0432 either way); (2) text-novelty Q1 strengthens when up/down removed (it was diluting the signal); (3) several v3 dims were partly up/down-driven and evaporate: `dim_info_type_supergroup` market_data +0.0083 (t+0.99, 10.9M) → −0.0122 (t−0.46, 1.4M); `dim_subject_specificity` 1-subject +0.0367 (t+7.47) → +0.0153 (n.s.); `dim_contract_horizon` <1h drops 99% in size. Because v6 excludes up/down from the primary view, the per-dimension tables above already reflect the excluded numbers.

Appendix C — The 50+ event_slug priors slice is one event family

All top-10 contracts in `dim_prior_settlements_bin__event_slug = '50+'` (only ~154 trade-bearing contracts mid-life) are "US strikes Iran by [DATE]" markets. Its spread reflects FLB on one recurring betting line, not a general "many priors → reversed FLB" pattern. Interpret as a single-event finding.

Appendix D — Multiple-testing correction

Family ≈ 3 windows × ~22 dims × ~3–5 kept slices × 2 weightings. Bonferroni-equivalent |t| at α=0.05: * > 2.5, ** > 3.0, *** > 3.5 (used throughout; the naive 1.96/2.58/3.29 are not). Of the audit's 87 v4 tests, 33 survived Bonferroni; all four headline findings pass with |t| > 12.

Appendix E — Known caveats

1. 3-way SE on small slices (Singleton, 50+ priors event_slug, N≈150) is asymptotically unstable; interpret those t-stats conservatively.
2. Lifecycle window non-comparability across durations: 25-80% is minutes for a <1h crypto but weeks for a >1mo election. Within `>1mo` the text-novelty Q1 spread is +0.0666 (t+21.5); within `<1h` it is n.s.
3. Dimension assignments use full-trade volume tiers, while calibration is on clean trades; the ~1.8% volume change does not move tier boundaries materially.
4. Dollar-weighted SE has fewer effective clusters than count-weighted (mass concentrates on large trades), so dollar t-stats run lower at equal spread; a dollar result that stays significant is therefore strong evidence.
5. **`dim_text_novelty`** bin choice: fixed semantic thresholds (`<0.50`, `0.50-0.75`, `0.75-0.90`, `0.90-0.95`, `>0.95`); the genuinely-isolated `<0.50` slice carries the headline FLB. Because up/down is excluded from the primary view, these are the up/down-excluded fixed-threshold numbers (the audit's recommended follow-up).
6. Closing window (80-100%) includes settlement-print artifacts; closing-line FLB on large/recurring slices may partly be mechanical close-out, not learning failure.
7. Offset vs slope: a uniform nonzero level (Sports/Esports) is a calibration offset, not FLB; the D10−D1 spread reported here is ~0 for those categories. The offset and its two-regime (mature-win/close-lose) structure are analyzed in `data_exploration.md`.